Chapter 9

Electron Microscopy Techniques for Imaging and Analysis of Nanoparticles

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Chapter Outline

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9.1 SCANNING ELECTRON MICROSCOPY

9.1.1 Instrumentation

Scanning electron microscopy (SEM) is probably the most common technique for analysis of particles. A modern SEM can furnish a resolution of 1 nm, relatively simple image interpretation, and large depth of focus, making it possible to directly image the three-dimensional structure of nanomaterials. SEM is a versatile tool that provides not only the morphological information of particles, but also chemical and in some cases crystallographic information.

An SEM is composed of an electron gun that provides an electron probe with high intensity, a condenser lens system that generates a small probe size, an objective lens for imaging, electron probe scanning coils, and a detector system for receiving images and spectra. The electron beam is formed by accelerating electrons generated from a filament set in a cathode held at a slightly negative or ground potential toward an anode whose potential is selectable. The cathode, filament, and anode are collectively referred to as the electron gun; the type, mode of operation, and cost of the filament depend on the brightness desired, its lifetime, and the vacuum level achievable in the electron gun chamber. The brightness β is defined as the current density per solid angle and can be calculated as

$$\beta = \frac{4I}{(\pi d_c \alpha_0)^2} \tag{9.1}$$

where I is the beam current, d_c is the diameter of the beam at its crossover between the cathode and the anode, and α_0 is the divergence semi-angle of the beam as it emerges from crossover. Tungsten hairpin filaments which operate by thermionic emission (i.e., filament heating) are the least expensive and can tolerate a relatively high pressure ($\sim 10^{-5}$ Torr), but they also produce relatively low brightness ($\sim 5 \times 10^4$ A/cm^2-ster) over a life time of $\sim$100 h. LaB_6 thermionic filaments have a typical lifetime of several hundred hours and are able to produce a higher brightness electron beam ($\sim 5 \times 10^5$ A/cm^2-ster) than tungsten thermionic filaments because of their lower work function, but they are also more expensive and require an electron gun vacuum of about 10^{-6} Torr.

Field emission sources can produce a brightness about two to three orders of magnitude greater than a LaB^6 filament and if handled with care can have a lifetime that exceeds that of LaB_6 filaments, but they are expensive and require a vacuum that is at least in the low 10^{-9} Torr range. However, if very high spatial resolution is a must, a field emission source is recommended because the electron tunneling that occurs through the few atoms at its sharp tip can produce an effective electron source size that is ~1 nm in diameter, which is considerably smaller than that of any of the thermionic electron sources. Moreover, the energy distribution of electrons generated by a field emission source is usually smaller than that of thermionic sources, which contributes to the production of a smaller source size by reducing the effect of condenser lens chromatic aberration.

Selection of the operating voltage and current depends on the nature of the sample and the information desired. Most modern SEMs can be run using an operating voltage in the range 100 V–30 kV. For small particles, usually an accelerating voltage of 10 kV or less is used since higher accelerating voltages have larger electron penetration depths into the sample and correspondingly larger electron probe interaction volumes with the sample.

Electron lenses are basically circular coils of wire whose axis coincides with the optic axis of the electron microscope. Application of a potential across the coil of wire produces a current in the coil; this in turn induces a magnetic field that has a significant component parallel to the optic axis. In addition, the objective lens contains a pole piece that serves to concentrate the magnetic field inside the lens. The pole piece has a gap, which will produce a fringing magnetic field with a small radial component. The magnetic field $\vec{B}$ that is the sum of the largely axial magnetic field produced by the current in the lens and the fringing field produced by the pole piece. Therefore, any electron in the beam that strays from the optic axis experiences a Lorentz force $\overline{F_L}$ that causes it to spiral toward the optic axis:

$$\vec{F_L} = -q(\bar{v} \times \bar{B}) \tag{9.2}$$

where q is the elementary charge and $\bar{v}$ is the velocity of the electron.

Electron lens design is complicated by the precision needed for a very small beam diameter and the addition of pole pieces, apertures, analytical detectors, etc. which break the symmetry of the simple circular wire coil design and introduce lens aberrations. Electron microscopes with smaller values of key aberration coefficients such as spherical aberration C_s and chromatic aberration C_c are preferred for work requiring high spatial resolution, but microscope cost also tends to scale inversely with the magnitudes of C_s and C_c.

Because C_s and C_c are set by the microscope manufacturer and often cannot be corrected by the operator, we will not delve into further details on electron lens design and electron optics effects on lens aberrations.

As its name implies, the function of the condenser lens and aperture is to demagnify or condense the beam diameter produced by the electron source.

Beam spatial coherency is affected by the choice of condenser aperture size, with a smaller aperture producing a more coherent beam (i.e., a more point-like source). While greater spatial coherency of the incident beam is desirable in imaging specimens requiring very high resolution, it is done so at the cost of some signal intensity loss from using a smaller condenser aperture.

The objective lens focuses the incident beam onto the sample surface; it is also where final correction of some beam aberrations (such as astigmatism) takes place. The distance over which the sample surface can be moved along the optic axis without degrading image resolution is called the depth of field, and the choice of objective aperture size affects the depth of field (Figure 9.1). In general, a smaller aperture produces a greater depth of field, but one must take into consideration that the image magnification M inversely affects the depth of field such that

$$\text{Depth of field} = \frac{d}{M\alpha} \tag{9.3}$$

where d is the lateral resolution of the image and α is the objective aperture semi-angle. In the case of imaging small particles where it is necessary to operate at or near the limit of the probe size $d_{\min}$, the optimum objective aperture semi-angle α_{opt} should be used, resulting in

$$\text{Depth of field at the instrument resolution} = \frac{\sqrt[3]{C_S d_{\min}^2}}{M} \tag{9.4}$$

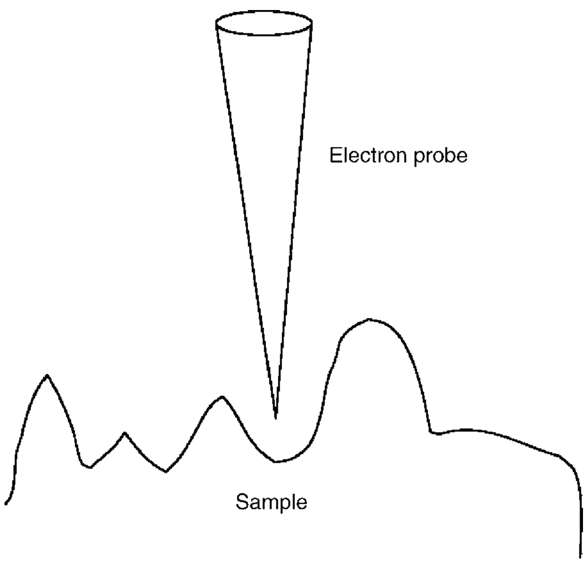

FIGURE 9.1 Schematic diagram showing the depth of field in scanning electron microscopy.

The working distance (i.e., the distance between the bottom of the final objective lens and the sample surface) may be adjusted slightly to achieve α_{opt}, but substantial changes in working distance may result in deviation away from the minimum probe size.[1]

Finally, as with all electron microscopes, vibration isolation and shielding from stray electromagnetic fields are critical if high spatial resolution is desired. Most electron microscopes are located on the lowest floor of a building away from vibration sources like elevators to reduce vibration problems. Commercially available vibration isolation solutions such as vibration isolation feet or vibration isolation platforms can be used satisfactorily with an electron microscope to help decouple it from vibrations transmitted through the floor. Some electron microscope facilities take the additional step of physically isolating the rooms in which high-resolution instruments are located by constructing those rooms on a foundation that is separate from the rest of the building. This may be necessary if the building is susceptible to low-frequency vibration from nearby sources such as railroad tracks, a truck route, or heavy construction. Microscope operators should also take care to eliminate high-frequency vibration sources (such as talking, touching the microscope, noise from nearby pumps, etc.) when taking high-resolution images. Strong fields generated by nearby pieces of equipment can produce beam instabilities and undesirable beam deflections in an electron microscope; shielding the exterior of the microscope column with a sheet of μ metal may be effective in stabilizing the beam. If stray fields continue to be problematic, more drastic measures such as isolating the microscope in its own shielded room and providing the microscope with its own electrical ground may be helpful.

9.1.1.1 Environmental Scanning Electron Microscope

The recent development of the environmental scanning electron microscope (ESEM) addresses the need to examine specimens under conditions as close as possible to their "natural" conditions, with a minimum of sample preparation. This includes the ability to image moist samples, which requires the ESEM to operate at relatively high pressures (up to ~50 Torr) using gases such as water vapor, nitrogen, argon, or carbon dioxide. The electron gun and most of the column of the ESEM are designed as for a conventional SEM, but in order to sustain such high pressures in the ESEM specimen chamber a series of apertures is used in the column of the ESEM just above the specimen chamber. Each aperture is designed to sustain a pressure difference of about 2–3 orders of magnitude that helps to minimize the scattering of electrons from the incident beam until the beam encounters the gas molecules in the specimen chamber. Collision of the incident electrons with the gas molecules causes beam spreading and therefore some degradation in the spatial resolution of ESEM images. However, the gas molecules also act as a secondary electron (SE) amplifier where SEs emitted from the sample ionize gas molecules and electrons released by the ionized gas molecules, can cause additional gas molecules to become ionized. These electrons are ultimately collected by a positively biased gaseous SE detector. The remaining positively charged gas ions in the specimen chamber help to neutralize any specimen charging that might occur, making the ESEM a useful tool for studying electrically insulating samples.

The spatial resolution of ESEM images is largely limited by the scattering experienced by the incident electrons when they collide with gas molecules in the specimen chamber. A spatial resolution of $\sim$5 nm may be achieved when operating the ESEM at the upper end of its pressure capabilities, with the resolution improving as the specimen chamber pressure decreases.

9.1.2 Signals produced by the SEM

9.1.2.1 Imaging

9.1.2.1.1 Backscattered Electron Imaging and Secondary Electron Imaging

For imaging SEM specimens, two main types of electrons emitted by the specimen are often used: backscattered electrons (BSEs) and SEs (Figure 9.2). In the case of imaging nanoscale materials, the choice of whether to use BSE or SE imaging depends mainly on the sample's topography sensitivity, composition sensitivity, and the orientation of the sample's small dimension (i.e., depth or lateral) with respect to the incident beam.

Backscattered electrons are relatively energetic electrons and are generated when incident probe electrons undergo elastic scattering by atoms in the sample through a total angle greater than 90°. A BSE can be generated from a single scattering event or from multiple scattering events, and the volume within the sample from which BSEs are produced generally increases with increasing accelerating voltage. Because BSE yield depends on the atomic number of the atom causing the backscattering, BSE images are sensitive to the elemental make-up of the sample.

Secondary electrons are conventionally regarded as having energies $<$50 eV and therefore the SEs that are detected are typically generated within a few nanometers of the sample surface. As such, SE images are sensitive to sample topography as well as surface chemical composition and surface electrical characteristics. SEs can be produced as the consequence of either single or multiple scattering events (i.e., produced directly from scattered incident electrons or produced from BSEs); the latter implies that the lateral extent from which SEs can be generated may be quite large.

The fact that SEs can be emitted either from the region covered by the incident electron probe and/or from an extended region as by-products of BSEs points to an important consideration in high-resolution imaging in the SEM: the image resolution depends not only on the incident probe diameter but also on the region of the specimen that is affected by the incident beam (either

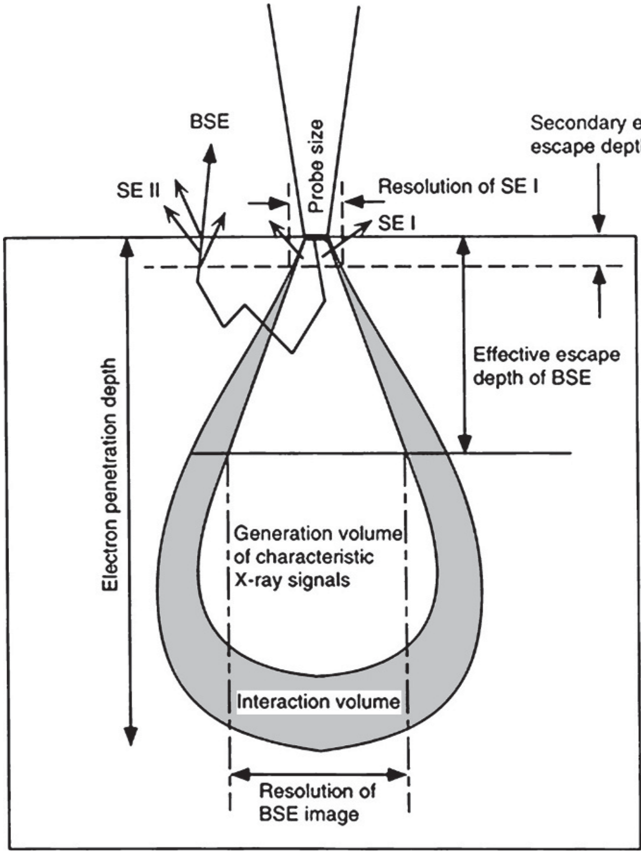

FIGURE 9.2 Schematic diagram showing the signals produced in beam-specimen interaction in scanning electron microscopy. SE I are secondary electrons produced from a single scattering event such as direct interaction between the incident electron beam and the sample. SE II are secondary electrons that are produced from a multiple scattering event such as from BSEs.

directly or indirectly), known as the beam–specimen *interaction volume*. The size and lateral extent of the interaction volumes that produce BSEs and SEs depend not only on the energy of the incident beam, but also the stopping power of the atoms contained in the sample. The interaction volume can be relatively large in polymer specimens, for example, because of the low atomic number of the target atoms. In BSE imaging, secondary factors that can affect the interaction volume include specimen thickness along the beam direction and specimen tilt with respect to the incident beam. The depth in the sample from which BSEs are emitted depends on the incident energy of the electron probe and the composition of the sample; greater amounts of high atomic number elements in a dense specimen provide more stopping power to the electron beam. The depth resolution of BSEs emitted by the sample can be improved by limiting the sample thickness and by tilting the sample so that incident electrons enter the specimen at a glancing angle with reduced depth penetration. The lateral extent from which BSEs are produced depends on the electron probe incident energy and the sample tilt with respect to the incident beam, with lower incident electron energies and normal incidence producing a smaller lateral region for BSE production.

These secondary factors of sample thickness along the incident beam direction and sample tilt tend not to have as great an impact in SE imaging. The SE interaction volume is in particular not strongly dependent on sample tilt; using a large sample tilt degrades the lateral spatial resolution of SE images because it results in a greater area of the sample surface intercepting the SE interaction volume and therefore a larger area from which SEs are emitted. Given these factors affecting the interaction volume, the imaging of small particles is typically carried out using SE imaging with a low operating voltage ($\sim$5 keV or less) to minimize the size of the interaction volume and to reduce the number of BSEs that might give rise to SEs.[1]

Since BSEs and SEs have different energies and are produced via different mechanisms, diverse information can be extracted by taking both a BSE image and a SE image from the same location of a sample. For example, unlike SEs, BSE production is strongly dependent on atomic number and therefore bright areas in a BSE image can reveal regions where higher concentrations of high atomic number elements reside. SE images are strongly influenced by specimen topography as well as any electric or magnetic fields that are present at or near the area of interest on the sample. Use of a low operating voltage and low current for the incident beam is advantageous in the case where the specimen is insulating, as these beam conditions help reduce specimen charging. Low operating voltage and low current are also recommended for samples that are beam-sensitive. In this case, tasks such as focusing and stigmating should be carried out not directly on the area of interest but in an adjacent, similar area; when the focus and astigmatism are set the specimen can be quickly moved to the area of interest for image recording. Use of a cooling sample holder may help reduce beam damage. A key trade-off in the case of beam-sensitive samples is that of incident electron dose received by the sample vs. the image signal-to-noise ratio. Some experimentation may be necessary to find the right combination of incident beam current and image scan speed that produces an acceptable image. If the sample is not susceptible to drift, several images of a given sample area taken at a high scan speed can be averaged together to improve the image signal-to-noise ratio.

A detector that is commonly used for sensing BSE and SE signals is the Everhart–Thornley detector, which basically consists of a scintillator surrounded by a Faraday cage attached to a light pipe and a photomultiplier. The scintillator converts the electron signal into light, and it is held at a potential of about +10 kV in order to attract electrons emitted from the specimen and to impart them with enough energy so that a sufficiently strong light signal can be produced when the electrons strike the scintillator. The light emitted by the scintillator is then guided by the light pipe to the photomultiplier, which amplifies the light signal for processing. By adjusting the bias on the Faraday cage, one can select whether BSEs or SEs enter the detector. Application of a small positive bias of a few hundred V to the Faraday cage serves to attract both BSEs and SEs to the detector. The magnitude of this bias is large enough so that the detector can attract some SEs that may originate outside of the direct line of vision between the detector and the specimen. By changing the bias on the Faraday cage to –50 V, SEs can be repelled by the detector without substantially altering the number of BSEs that enter the detector. The BSEs that do enter the detector when the Faraday cage is negatively biased in this way are from regions of the specimen that fall within the line of sight of the detector. Therefore this can give rise to a "directional" appearance and "shadows" in SEM images formed using only BSEs, whereas sample features in SEM images that include SEs may look as if they are more uniformly illuminated. Because of this directional effect on BSE collection, the Everhart–Thornley detector is generally used in SE imaging.

For BSE imaging in the SEM, solid state detectors consisting of a pair electrodes attached to opposite sides of a semiconductor p–n junction and a current amplifier are frequently used. As an energetic BSE strikes the electrode on the entry face of the detector, it creates a number of electron–hole pairs in the semiconductor that is directly proportional to its energy. The intrinsic electric field of the p–n junction helps to separate the electrons and the holes that are created, and the electrons are swept toward the current amplifier. The relatively low energies of SEs are not sufficient to create significant quantities of electron–hole pairs and therefore SEs are not influential in images obtained using a solid state detector. The active area of a solid state detector typically covers a large radius and a large range of polar angles with respect to the optic axis, which helps to reduce the directional effects produced by the Everhart–Thornley detector in BSE imaging.

Figure 9.3 shows an SEM image recorded from gold particles deposited on an alumina substrate using a template provided by a photonic crystal. The periodically arranged Au nanoparticles serve as the catalyst for growing nanorods/nanowires.[2] Figure 9.4 shows the SEM image of the ZnO nanorods grown from the gold nanoparticles. The spatial distribution of the nanorods is determined by the distribution of the Au catalyst.[3]

9.1.2.1.2 Low-Voltage Imaging

Low voltage, high-resolution SEM is an effective technique for imaging specimens that are sensitive to electron radiation. Low-voltage SEM has attracted a great deal of attention in recent years because of its application in polymers, biology, and materials science. The resolution of low-voltage SEM is primarily

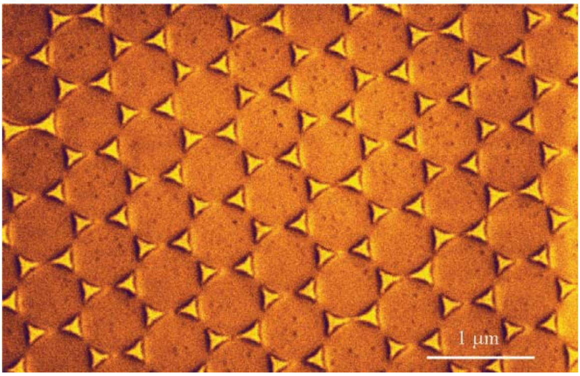

FIGURE 9.3 SE image of gold particles deposited on an alumina substrate using a template provided by a photonic crystal.

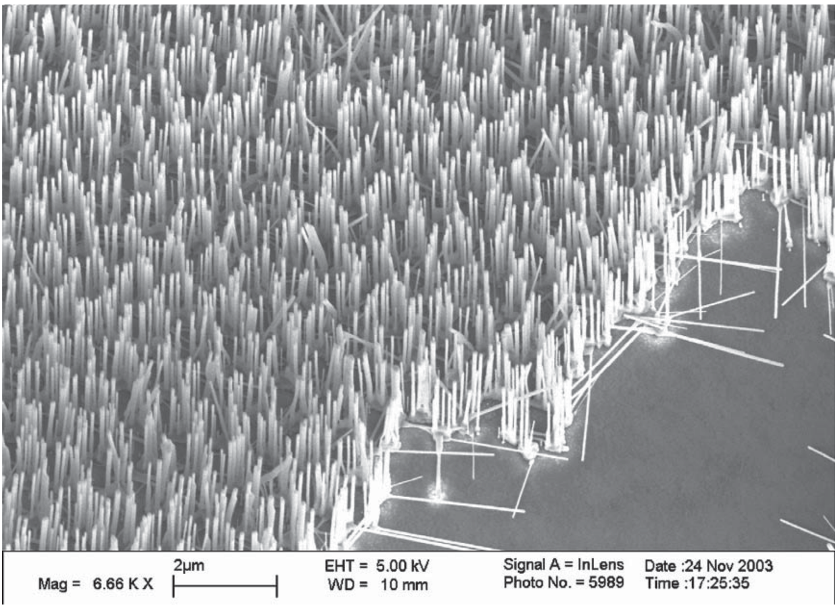

FIGURE 9.4 SE image of ZnO nanorods grown from gold nanoparticles.

limited by chromatic aberration. If the energy spread of the electron emission source is ΔE, the incident beam energy is E_0 and the chromatic aberration coefficient is C_c, the probe size limited by this effect is given by

$$d_c = \alpha \frac{\Delta E}{E_0} C_c \tag{9.5}$$

where α is the beam convergence angle. The resolution of an SEM at 200 V can be as high as 2.5 nm, which is capable of resolving many nanoscale defects. A low operating voltage (<5 kV) is also desirable in the case where the sample is an electrical insulator and application of a conductive coating to the sample is undesirable. In a low accelerating voltage regime, it is possible to achieve conditions where the electron emission from the sample exceeds the incident electrons from the electron beam, thereby preventing sample charging.[4]

9.1.2.2 Composition Analysis

9.1.2.2.1 X-Rays

There are a number of outputs that result from the interaction of incident probe electrons in the SEM with the atoms and electrons in the sample. Signals that are collected and amplified from these outputs are often the basis for obtaining analytical information about the sample.

X-rays are among the outputs emitted from the sample, and they are commonly used to determine the composition of the sample. X-rays can be produced when an energetic incident electron knocks out a core electron of an atom in the sample. This puts the atom in an excited state, and in order to achieve a state that is more energetically stable an electron from an outer electron shell of the atom will reduce its energy by occupying the core electron vacancy. The amount of energy reduction experienced by the atom is then equal to the difference between the energy levels of the outer shell and the core shell where the vacancy occurred, and following energy conservation principles an X-ray having this amount of energy is emitted. Because the energy difference between different electron shells is characteristic for each element, analysis of the energies of X-rays generated by the sample offers information about the composition of the sample. The energies of the emitted X-rays are usually on the order of a few keV, which means that both incident probe electrons and sufficiently energetic BSEs can give rise to X-ray production. This implies that the volume in the sample from which X-rays are emitted can be relatively large in comparison to the probe diameter; this volume tends to increase with increasing incident electron energy and decreasing sample density. The volume from which X-rays are produced from a sample in SEM is frequently larger than the probe dimensions, making chemical analysis of small particles in the SEM quite challenging.

One problem associated with chemical analysis of small particles is the extraneous signal associated with material other than the particles themselves (e.g., the signal from a supporting substrate). If the approximate composition of the particles is known, a substrate material that does not have any of the elements contained in the particles and that is stable under an electron beam should be used; the signal from the substrate is then easily identified and removed from chemical analysis. Lowering the operating voltage also helps to reduce the volume in the specimen from which X-rays are produced. If the composition of the small particles is unknown, a first step for composition analysis using the SEM is to deposit the particles onto a transmission electron microscope grid covered with a thin carbon film. This film consists mostly of carbon and is usually thin enough so as not to make a significant contribution to the X-ray data.

The energies of emitted X-rays are usually analyzed by at least one of two ways: wavelength dispersive spectrometry (WDS) or energy dispersive spectrometry (EDS). In a WDS spectrometer, X-rays emitted by the sample strike an analyzing crystal with known interplanar spacing d_{hkl}. By choosing an analyzing crystal with an appropriate d_{hkl}, the Bragg relation

$$n\lambda = 2d_{hkl}\sin\theta,\ \ n\,\text{a positive integer} \tag{9.6}$$

can be satisfied by rotating the analyzing crystal so that X-rays with wavelength λ strike the $\{h \cdot k \cdot l\}$ planes of the crystal at the Bragg angle θ_B. Therefore, X-rays with energy E are diffracted by the analyzing crystal into an amplifier and counter, where the energy E of the X-ray is related to its wavelength λ at typical SEM accelerating voltages (i.e. where relativistic effects are negligible) by

$$E = \frac{h^2}{2m_e\lambda^2} \tag{9.7}$$

where h is the Planck's constant and m_e is the rest mass of the electron. The diffracted X-ray signal is amplified when it enters a proportional counting tube, which contains a biased wire and is typically filled with an inert gas. When the gas atoms absorb the incoming X-rays, they become excited and emit photoelectrons that are attracted to the biased wire. As they photoelectrons proceed toward the biased wire, they can scatter elastically with electrons in other gas atoms it encounters along the way to produce a thousand-fold or greater amplification in signal. This structure of the WDS detector highlights its key strength as well as a drawback: good signal-to-noise and resolution can be achieved using WDS, but signal production using Bragg diffraction from a crystal analyzer implies that WDS is able to detect only one characteristic X-ray peak at any given time, making a search for multiple elements in a sample using WDS potentially very time consuming.

The EDS is also commonly used to analyze X-rays emitted by a sample in the SEM. Here, the X-ray enters the EDS spectrometer through a very thin window made of low atomic number material. The thickness and composition of this window may absorb low-energy X-rays and therefore determine the lower limit on the atomic number of the elements that can be detected by the spectrometer. The X-ray then strikes a detector consisting of a reverse biased, lithium-doped silicon p–i–n junction (p-type, intrinsic, n-type), creating electron–hole pairs in the intrinsic region of the junction. Since it takes about 3.8 eV to create an electron–hole pair, the number of electron–hole pairs created is proportional to the X-ray energy. The reverse bias applied to this junction helps to separate the electrons and the holes that are created, and the electrons are swept toward the current amplifier. The detector is kept cool with liquid nitrogen to prevent the lithium from diffusing away from the intrinsic region of the junction and to reduce noise. EDS is favored in the analysis of X-rays emitted by SEM samples largely because it can detect the characteristic X-ray peaks of many elements simultaneously, but WDS may be more useful in cases where high resolution of X-ray energies is required and/or very low atomic number elements need to be detected.

In analyzing the composition of nanostructures, it may be difficult to obtain sufficient signal from the sample. If the sample is stable under the electron beam and contamination is not an issue, a long data acquisition time may be necessary for an acceptable X-ray spectrum. Otherwise, the data can be improved by adding together the X-ray spectra from compositionally identical nanostructures to effectively create a single X-ray spectrum whose intensity and acquisition time is equal to the sum of the signals and acquisition times of the component spectra.

When performing quantitative analysis of SEM X-ray data, it is best to have X-ray data from a standard specimen for comparison. However, this is often not possible in the case of new materials development. Care must be exercised in the analysis of X-ray spectra, as there are some elements that share similar characteristic X-ray energies, and in quantitative analysis of X-ray spectra the relative intensities of the X-ray peaks of elements do not necessarily correspond to the relative amounts of those elements present in the sample. Common sample-based corrections that need to be taken into account are atomic number (Z) effects, absorption (A) effects, and fluorescence (F) effects; these three factors are collectively referred to as the *ZAF* correction. The atomic number factor takes into consideration the increased probability that interaction with an incident electron will produce BSEs rather than X-rays, and that the electron stopping power (which in inversely proportional to the density of the sample) decreases with increasing Z. Absorption accounts for the increased probability that an X-ray emerging from an atom may be absorbed by another atom in the sample as its path length inside the sample increases; this depends largely upon the incident electron energy and the specimen tilt with respect to the X-ray detector. In the phenomenon of fluorescence, X-rays with sufficient energy can cause emission of characteristic X-rays from a nearby atom through the same excitation—relaxation process by which an incident electron produces X-rays from the specimen.

In addition, there are detector-based corrections that need to be taken into account in X-ray spectrum analysis. An example of a detector-based correction is the type and thickness of the X-ray detector window and any ice build-up on the X-ray detector window that can cause absorption of lower energy X-rays. Another correction is the accounting of extraneous X-ray peaks in the spectrum resulting from sufficiently energetic X-rays entering the detector and causing the fluorescence of characteristic X-rays from a detector component, such as from the gas in the proportional counting tube (in WDS) or from the silicon in the p–i–n junction (in EDS). This detector-based fluorescence peak is often accompanied by an *escape peak* whose energy is equal to the difference between the energy of the X-ray entering the detector and the energy of the detector-based fluorescence peak. Multiple X-rays that enter the detector within the time resolution of the detector's pulse processor may be processed as a false single X-ray called a *sum peak* whose energy is equal to the sum of the energies of the contributing X-rays.

User-induced factors can also complicate the interpretation of X-ray spectra. Contamination of the sample through improper handling or storage can give rise to peaks in the X-ray spectrum from the contaminant material (usually hydrocarbons). If the sample is able to tolerate it, vacuum baking or ion milling may help remove surface contamination prior to examination in the SEM. Use of a cold finger in the SEM can help reduce the effects of contamination by providing a surface that is colder than the sample where contaminant molecules can condense.

9.1.2.2.2 Auger Electrons

When an atom in the specimen relaxes from the excitation caused by an incident electron producing an inner shell vacancy, it can emit excess energy in one of several forms. In the previous section we described the consequences when this excess energy is released in the form of an X-ray photon. However, this excess energy need not always be emitted in the form of an X-ray photon; through a similar de-excitation process an atom can also emit an electron known as an *Auger electron* that is also characteristic of each element and of the electron shells involved in the relaxation process. An atom cannot emit both a photon and an Auger electron when it relaxes from a given outer shell electron filling a particular inner shell vacancy. While Auger electron spectroscopy (AES) is not commonly used with the SEM, it may be advantageous to use the Auger electron signal in cases where the X-ray yield is known to be low (such as in low atomic number elements).

Probably the main reason why AES is not often used is because Auger electrons are typically generated from within $\sim$10 Å of the specimen surface. For accurate AES analysis, this necessitates ultra-high vacuum (UHV) conditions in SEM and very careful sample preparation and handling to avoid the formation of unwanted surface layers on the sample. Like SEs, Auger electrons are generated very close to the specimen surface and hence the factors that limit SE image resolution also apply in determining the lateral resolution of Auger electron spectra.

9.2 HIGH-RESOLUTION TRANSMISSION ELECTRON MICROSCOPY

High-resolution transmission electron microscopy is the most powerful technique for direct imaging of atoms in nanoparticles. A conventional TEM usually has an image resolution of better than 0.2 nm and is very capable of resolving atomic scale structures of nanoparticles. The most important development in recent years is the integration of HRTEM imaging with high spatial resolution chemical analysis. By forming a nanometer size electron probe, as small as 0.2 nm, TEM is unique in identifying and quantifying the chemical and electronic structures of individual nanoparticles using electron energy-loss spectroscopy (EELS) and EDS. This is the high-spatial resolution spectroscopy.

9.2.1 Main Components of a Transmission Electron Microscope

Both the TEM and the SEM share the same electron gun design (Section 9.1.1). A schematic of a TEM is illustrated in Figure 9.5. Most modern TEM electron guns use either a LaB_6 filament or a field emission filament to obtain the desired brightness. The condenser lens also serves the same function in both the SEM and the TEM and therefore TEM condenser lenses are designed in the same way as SEM condenser lenses.

In addition to focusing the incident beam onto the specimen surface, the objective lens [and the intermediate lens(es)] in a TEM also provides some magnification. The magnification ability of an objective lens is determined in part by a pole piece situated within the objective lens, where the pole piece provides an additional magnetic field concentrated near the specimen. The specimen is usually placed within the objective lens during TEM study, and the relative

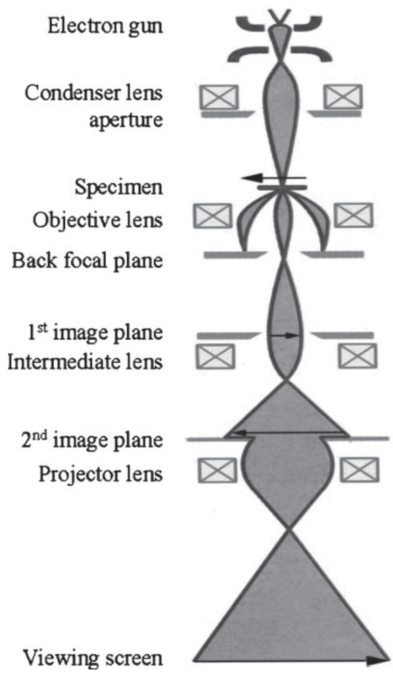

FIGURE 9.5 Schematic optical diagram of a transmission electron microscope.

position of the specimen and the objective aperture gives rise to different imaging modes that can be used in the TEM. Specifically, the incident electron beam passes through the specimen before reaching the objective aperture, which means that the objective aperture can be used to select the electrons scattered by the specimen used in forming an image. If the objective aperture is positioned to include the incident beam that is unscattered by the thin TEM sample, the resulting image is known as a *bright field* (BF) image. If the objective aperture excludes the unscattered portion of the incident beam, the resulting image is called a *dark field* (DF) image. By including the unscattered portion of the incident beam, one of the primary contrast mechanisms in BF images is mass thickness contrast. Contrast mechanisms that are important in DF images include diffraction contrast and contrast due to defects and/or deviation from crystallinity in the sample. An electron detector (usually in the form of a phosphor screen, a CCD camera, or electron-sensitive film) is placed in the image plane for viewing the resulting image or scattering pattern.

A selected area diffraction (SAD) aperture and a projector lens located between the objective lens and the image plane are used for viewing diffraction patterns. With the TEM operating in a normal BF or DF imaging mode, the size of the SAD aperture is chosen if diffraction pattern information is desired only from a specific location on the specimen. When the TEM is switched to diffraction mode, electrons that have been scattered through the same angle by the sample are focused to a point in the back focal plane of the TEM, and the projector lens serves to project an image of the back focal plane onto the image plane of the TEM to create a viewable diffraction pattern. The formation of a diffraction pattern on the image plane is typically accompanied by the use of a relatively large objective aperture and a reduction in strength in the intermediate lens(es).

Analytical equipment such as an EDS detector or and electron energy-loss spectroscopy (EELS) can also be found on some TEMs.

9.2.2 The Physics for Atomic Resolution Lattice Imaging

9.2.2.1 Phase Contrast Imaging

Images in TEM are usually dominated by three types of contrast. First, diffraction contrast,[5] which is produced due to a local distortion in the orientation of the crystal (by dislocations, for example), so that the diffracted intensity of the incident electron beam is perturbed, leads to contrast observed in bright-field image. The nanocrystals (NC) oriented with their low-index zone-axis parallel or nearly parallel to the incident beam direction usually exhibit dark contrast in the BF image that is formed by selecting the central transmitted beam. Since the diffraction intensities of the Bragg reflected beams are strongly related to the crystal orientations, this type of image is ideally suited for imaging defects and dislocations. For NC, most of the grains are defect-free in volume, while a high density of defects are localized at the surface or grain boundary. Thus, diffraction contrast can be useful for capturing strain distribution in NC whose sizes are larger than 15 nm. For smaller size NC, since the resolution of diffraction contrast is in the order of 1–2 nm, its application is limited.

Phase contrast is produced by the phase modulation of the incident electron wave that transmits through a crystal potential,[5] which is the dominant mechanism for high-resolution imaging. This type of contrast is sensitive to the atom distribution in the specimen and it is the basis of high-resolution TEM. To illustrate the physics of phase contrast, we consider the modulation of a crystal potential to the electron wavelength. To show the phase contrast imaging, we use a very thin sample [Figure 9.6(a)], and consider the phase shift to the electron wave after transmission. When the electron goes through a crystal potential field, its kinetic energy is perturbed by the variation of the potential field, resulting in a phase shift with respect to the electron wave that travels in a space free of potential field. For a specimen of thickness d, the phase shift ϕ is approximately

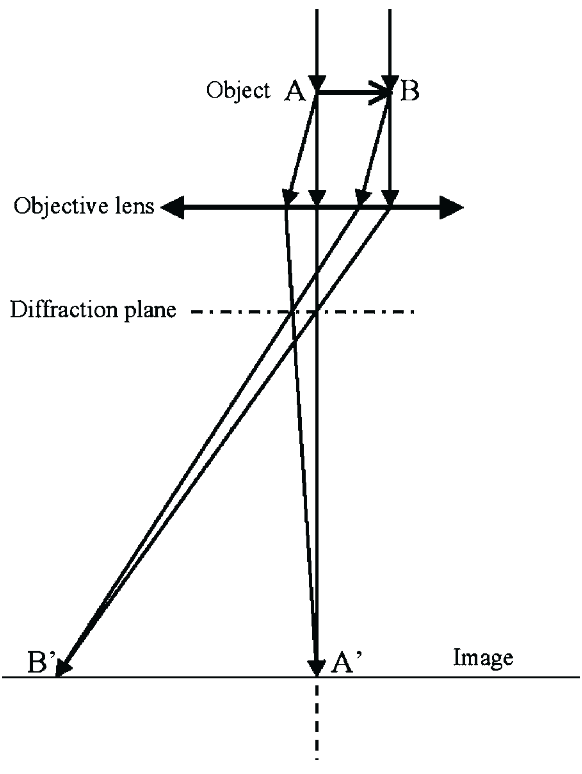

FIGURE 9.6 Abbe's theory of image formation in a one-lens transmission electron microscope.

$$\o \approx \sigma V_p(b) = \sigma \int_0^d \mathrm{d}z V(r) \tag{9.8}$$

where $\sigma = \pi/\lambda U_0$, (x, y) are the lateral coordinates perpendicular to the incident beam direction, U_0 is the acceleration voltage, and $V_p(x, y)$ the thickness-projected potential of the sample. Therefore, from the phase point of view, the electron wave is modulated by a phase factor

$$\Psi(x, y) = \exp\left[\mathrm{i}\sigma V_p(x, y)\right] \tag{9.9}$$

This is known to be the *phase object approximation* (POA), in which the sample acts as a phase grating filter, and its atomic distribution projected along the beam direction is directly captured in the electron phase [Figure 9.6(b)]. If the incident beam travels along a low-index zone-axis, the variation of V_p (**b**) across atom rows is a sharp function because an atom can be approximated by a narrow potential well and its width is on the order of 0.2–0.3 Å. This sharp phase variation is the basis of *phase contrast*, is fundamental in atomic-resolution imaging in TEM.

Finally, in mass-thickness or atomic number produced contrast, atoms with different atomic numbers exhibit different powers of scattering. If the image is formed by collecting the electrons scattered to high-angles, the image contrast would be sensitive to the average atomic number along the beam direction. This type of contrast is less useful for analysis of nanoparticles.

9.2.2.2 Image Formation

As introduced above, phase contrast is fundamental in high-resolution imaging. We now introduce the image formation process that is about the information transfer of the electron phase through the optical system in TEM. Image formation in TEM is analogous to that in an optical microscope.[6] For easy illustration, a TEM is simplified into a single lens microscope, as shown in Figure 9.6, in which only a single objective lens is considered for imaging and the intermediate lenses and projection lenses are omitted. This is because the resolution of the TEM is mainly determined by the objective lens. The entrance surface of a thin foil specimen is illuminated by a parallel or nearly parallel electron beam. The electron beam is diffracted by the lattice planes of the crystal, forming the diffracted beams which are propagating along different directions. The electron–specimen interaction results in phase and amplitude changes in the electron wave that are determined by quantum mechanical diffraction theory. For a thin specimen and high-energy electrons, the transmitted wave function $Y(x, y)$ at the exit face of the specimen can be assumed to be composed of a forward-scattered wave.

The propagation through the objective lens is the main source of nonlinear information transfer in TEM. The diffracted beams will be focused in the back-focal plane, where an objective aperture could be applied. An ideal thin lens brings the parallel transmitted waves to a focus in the back focal plane. Waves leaving the specimen in the same direction (or angle θ with the optic axis) are brought together at a point on the back focal plane, forming a diffraction pattern. The electrons scattered to angle θ experience a phase shift introduced by the chromatic and spherical aberrations of the lens, and this phase shift is a function of the scattering angle, thus, the diffraction amplitude at the back-focal plane is modified by

$$\psi'(\mathbf{u}) = \psi(\mathbf{u}) \exp[i\chi(\mathbf{u})] \tag{9.10}$$

where $\psi(\mathbf{u})$ is the Fourier transform of the wave $\Psi(\mathbf{r})$ at the exit face of the specimen, $\mathbf{u}$ is the reciprocal space vector that is related to the scattering angle by $u = 2\sin\theta/\lambda$, and $\chi(\mathbf{u})$ is determined by the spherical aberration coefficient C_s of the objective lens and the lens defocus Δf

$$\chi(\mathbf{u}) = \frac{\pi}{2} C_S \lambda^3 u^4 - \pi \Delta f \lambda u^2 \tag{9.11}$$

where λ is the electron wavelength. The $\exp[i\chi(\mathbf{u})]$ represents the distortion of the wave by the non-linear information transfer characteristic of the optical system. The chromatic aberration arises from the change in focal length as a result in energy fluctuation of the incident electron beam. The spherical aberration is due to the dependence of the focal length on the electron scattering angle θ. From simple optics, a change in focal length changes the phase of the electron beam. Thus, the aberration and defocus of the lens modulate the phases of the Bragg beams scattered to different angles.

In the electron diffraction plane, there are many beams that have been scattered by the sample and they carry the phase and amplitude information from the sample. The electron image is the result of interference among the beams scattered to different angles, and this interference pattern is affected by the phase modulation introduced by the aberrations of the objective lens. The image is calculated according to

$$I(x, y) = |\Psi(x, y) \otimes t_{\text{obj}}(x, y)|^2 \tag{9.12}$$

where $\otimes$ indicates a convolution calculation of (x, y), $t_{\text{obj}}(x, y)$ is the inverse Fourier transform of the phase function $\exp[i\chi(\mathbf{u})]$. The convolution of the lens transfer function introduces the non-linear information transfer characteristics of the objective lens, leading to complexity in image interpretation. It is apparent that the phase information is contained in the image, and the problem is how to interpret the image.

9.2.2.3 Image Interpretation of Very Thin Samples

Image interpretation is very important in TEM. In a HRTEM image, one usually wonders if the atoms are in darker or brighter contrast. To answer this question, we adopt the *weak scattering object approximation* (WPOA), which assumes that the scattering power of the sample is so weak so that the phase modulation is small. If the specimen is so thin that the projected potential satisfies $|\sigma V_p(x, y)| \ll 1$, from Eq. (9.9), the phase grating function is approximated by

$$\Psi(x, y) \approx 1 + i\sigma V_p(x, y) \tag{9.13}$$

From Eq. (9.12) and ignoring the σ^2 term, the image intensity is calculated by

$$I(x, y) \approx 1 - 2\sigma V_p(x, y) \otimes t_s(x, y) \tag{9.14}$$

where $t_s(x, y) = \mathrm{Im}[t_{\mathrm{obj}}(x, y)]$. The second term in Eq. (9.14) is the interference result of the central transmitted beam with the Bragg reflected beams. Any phase modulation introduced by the lens would result in contrast variation in the observed image.

To reveal the true atomic scale structure of the sample, the experimental condition has to be chosen in such a way that the $t_s(x, y)$ function is a sharp peak function and it can be approximated by a delta function. It has been found that by choosing the defocus to be $\Delta f = [4/3 C_s \lambda]^{1/2}$, which is known as the Scherzer defocus, $t_s(x, y)$ is approximated to be a negative Gaussian-like function with a small oscillating tail. Thus, the image contrast using the WPOA is directly related to the two-dimensional thickness-projected potential of the crystal, and the image reflects the projected structure of the crystal and the atoms show dark contrast. This is the basis of structure analysis using high-resolution TEM. On the other hand, the contrast of the atom rows is determined by the sign and real space distribution of $t_s(x, y)$. Therefore, the image contrast in HRTEM is critically affected by the defocus value. A change in defocus could lead to contrast reversal, which makes it difficult to match the observed image directly with the projection of atom rows in the crystal. This is one of the reasons that image simulation is a key step in quantitative analysis of HRTEM images.

It must be pointed out that the WPOA is probably the simplest model for illustrating the physics in image formation, but it is not valid for a general case because the sample is usually thicker than we would like it to be. Quantitative analysis of high-resolution TEM image requires numerical calculations using full dynamic theory for electron scattering.

9.2.2.4 Dynamic Theory and Image Simulation

In practice, interaction between the incident electron and the specimen is rather complex and multiple scattering is inevitable. We usually need to use rigorous dynamic theory for image simulation. Here we introduce the fundamentals of image simulation. Image simulation needs to include two important processes: the dynamic multiple scattering of the electron in the crystal and the information transfer of the objective lens system. The dynamic diffraction process is to solve the Schrödinger equation under given boundary conditions.

There are several approaches for performing dynamic calculations,[6] among which the multislice theory[7] and the Bloch wave theory are the most frequently used ones. For a finite size crystal containing defects and surfaces, the multislice theory is most adequate for numerical calculations. The multislice many-beam dynamic diffraction theory was first developed based on the physical optics approach. The crystal is cut into many slices of equal thickness Δz normal to the direction of the incident beam (Figure 9.7). When the slice thickness tends to be very small the scattering of each slice can be approximated as a phase object, the transmission of the electron wave through each slice can be considered separately if the backscattering effect is negligible, which means that the calculation can be made slice-by-slice. The defect and 3D crystal shape can be easily accounted for in this approach.

The transmission of the electron wave through a slice can be considered as a two step process—the phase modulation of the wave by the projected atomic potential within the slice and the propagation of the modulated wave in "vacuum" for a distance Δz along the beam direction before striking the next crystal slice. The wave function before and after transmitting a crystal slice is correlated by

$$\Psi(x,y,z+\Delta z) = [\Psi(x,y,z)Q(x,y,z+\Delta z)] \otimes P(x,y,\Delta z) \tag{9.15}$$

where the phase grating function of the slice is

$$Q(x,y,z+\Delta z) = \exp\left[i\sigma \int_{z}^{z+\Delta z} dz V(b,z)\right] \tag{9.16a}$$

which is given to account for the phase modulation introduced by the thin crystal slice. The propagation function is

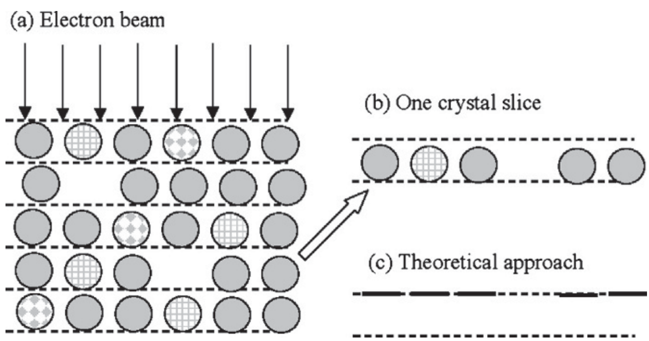

FIGURE 9.7 (a) A schematic diagram showing the physical approach of the multislice theory for image and diffraction pattern calculations in TEM. (b) Transmission of electron wave through a thin crystal slice. (c) An approximate treatment of the wave transmission through a thin slice.

$$P(b, \Delta z) = \frac{\exp\left(piK|b|^2/\Delta z\right)}{i\lambda\Delta z} \tag{9.16b}$$

which represents the Fresnel propagation of the electron for a distance Δz in vacuum.

The most important characteristic of this equation is that no assumption was made regarding the arrangement of atoms in the slices, so that the theory can be applied to calculation of the electron scattering in crystals containing defects and dislocations. This is the most powerful approach for NC.

The next step is the information transfer through the objective lens system. By taking a Fourier transform of the exit wave function, the diffraction amplitude is multiplied by the lens transfer function $\exp(i\chi)\,E(\mathrm{u})$, where the defocus is a variable. The defocus value of the objective lens and the specimen thickness are two important parameters which can be adjusted to match the calculated images with the observed ones.

Figure 9.8 shows systematic simulations of a decahedral Au particle in different orientations. The particle shape can only be easily identified if the image is recorded along the five-fold axis. The group A and group B images were calculated for two different defoci, exhibiting contrast reversal from dark atoms to bright atoms. In practice, with consideration of the effects from the carbon substrate, it is difficult to identify the particle shape if the particle orientation is off the five-fold axis.

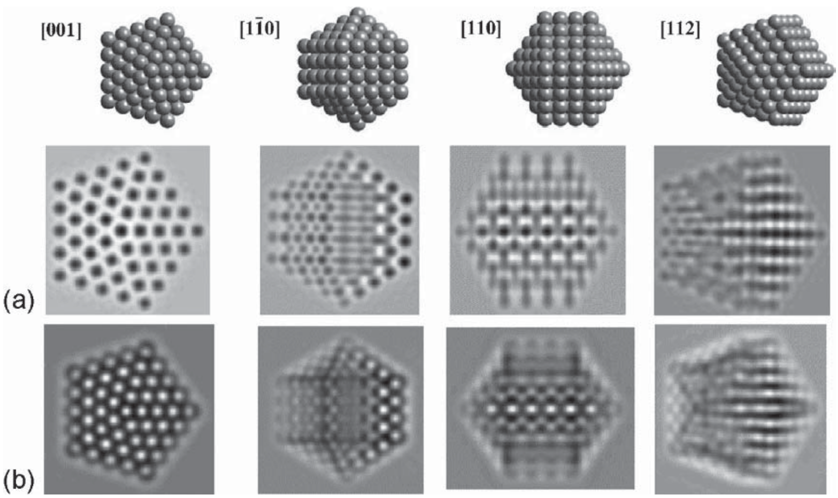

FIGURE 9.8 Simulated images for a decahedral Au particle at various orientations and at foci of (a) $\Delta f = 42$ nm and (b) $\Delta f = 70$ nm. The Fourier transform of the image is also displayed. *(Courtesy of Drs. Ascencio and M. José-Yacaman.)*

9.3 SHAPES OF NANOCRYSTALS

9.3.1 Polyhedral Shapes

Nanocrystals exhibiting distinctly different properties from bulk are mainly due to their large portions of surface atoms, and the size effect and possibly the shape effect as well. Since different energy exhibited by different crystallographic planes, a nanoparticle constituting a finite number of atoms is bounded by some lower energy crystal planes, thus it has a specific geometrical shape. Surface energies associated with different crystallographic planes are usually different. Take face-centered cubic structured metal as an example: a general sequence may hold for surface energy of, $\gamma\{111\}<\gamma\{100\}<\gamma\{110\}$. Facets tend to form on the particle surface to increase the portion of the low-index planes. Therefore, for particles smaller than 10–20 nm, the surface is a polyhedron.

Figure 9.9(a) shows a group of cubo-octahedral shapes as a function of the ratio, R, of the growth rate of the $\langle 100\rangle$ to that of the $\langle 111\rangle$.[8] The fastest growth direction in a cube is $\langle 111\rangle$, the longest direction in the octahedron is the $\langle 100\rangle$ diagonal, and the longest direction in the cubo-octahedron ($R=0.87$) is the $\langle 110\rangle$ direction. The particles with $0.87<R<1.73$ have $\{100\}$ and $\{111\}$ facets, which are named the truncated octahedral (TO). The other group of particles has a fixed (111) base with exposed $\{111\}$ and $\{100\}$ facets [Figure 9.9(b)]. An increase in the area ratio of $\{111\}$ to $\{100\}$ results in the evolution of particle shapes from a triangle-based pyramid to a tetrahedron.

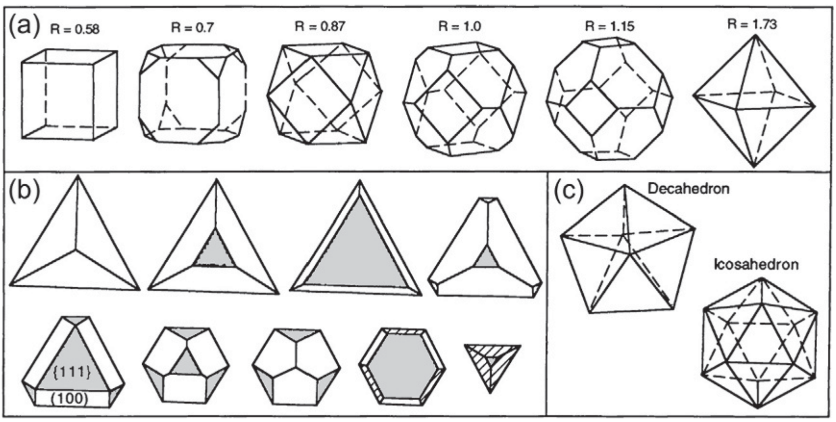

FIGURE 9.9 (a) Geometrical shapes of cubo-octahedral nanocrystals as a function of the ratio, R, of the growth rate along the $\langle 100\rangle$ to that of the $\langle 111\rangle$. (b) Evolution in shapes of a series of (111) based nanoparticles as the ratio of $\{111\}$ to $\{100\}$ increases. The beginning particle is bounded by three $\{100\}$ facets and a (111) base, while the final one is a $\{111\}$ bounded tetrahedron. (c) Geometrical shapes of multiply twinned decahedral and icosahedral particles.

Imaging of facets and atoms on the facets of NC relies on the profile imaging technique. If a particle is oriented along a low-index zone axis, the distribution of atoms on the surface can be imaged in profile, and the surface structure is directly seen with the full resolution power of a TEM.[9] This is a powerful technique for direct imaging of the projected shapes of nanoparticles particularly when the particle size is small. With consideration of the symmetry in particle shapes, HRTEM can be used to determine the 3D shape of small particles although the image is a 2D projection of a 3D object. It usually requires at least two images recorded from different orientations to define the particle shape.

We now take Co particles as an example to show the experimental details for particle shape determination.[10] Cobalt NC prepared from solution phase reduction of cobalt chloride in the presence of stabilizing agents has been determined by X-ray diffraction to exhibit the β-Mn structure (ε-Co phase, space group $P4_132$)[11] with a cubic unit cell of 20 atoms and 6.30 Å on a side. Figure 9.10 shows an electron diffraction pattern recorded from a Co NC assembly. This diffraction pattern is consistent with the X-ray diffraction data reported previously with the most intense diffraction peaks appearing as {221}, {310}, and {311}. The three rings were enclosed by the objective aperture of the HRTEM, thus the atomic-scale lattice images are dominated by the interference among the three beams and the central transmitted {000} beam.

The two-dimensionally projected shape of the Co nanocrystal is easily seen in the HRTEM images shown in Figure 9.11. From X-ray diffraction studies of the Co NC assembly, it is known that the (221) and (310) planes diffraction, but

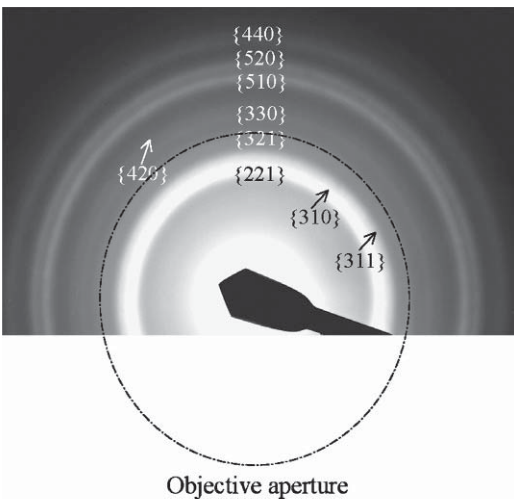

FIGURE 9.10 Electron diffraction pattern recorded from the as-synthesized Co nanocrystals, which matches well to the intensities and indices for the β-Mn structure.

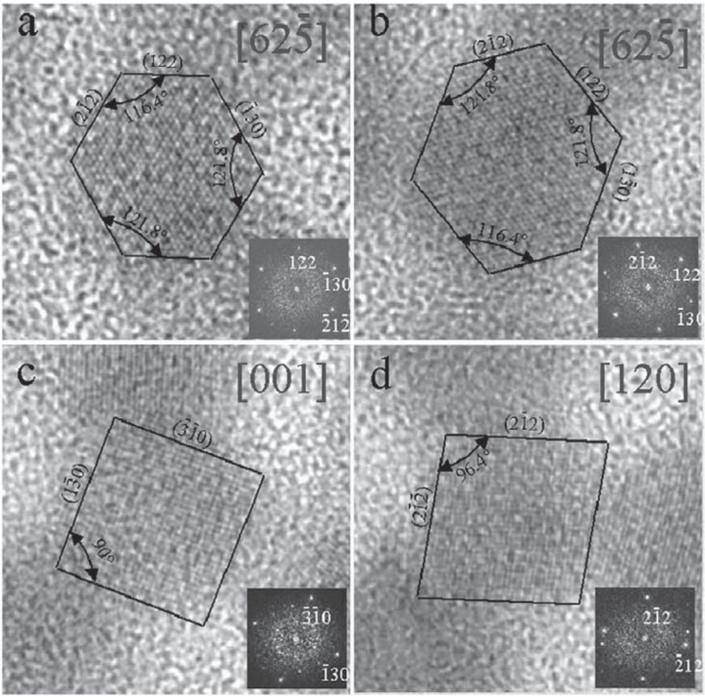

FIGURE 9.11 Typical HRTEM images of the Co nanocrystals oriented along (a, b) [62 $\bar{5}$], (c) [001] and (d) [120], showing the {221} and {310} facets. The inserts are the Fourier transforms of the corresponding images.

the interpretation of the image has to consider the structure of the ε-Co phase. From the electron diffraction pattern, the dominant diffraction that will contribute to the images will be the {221} and {310} reflections with the central transmitted beam. By matching the interplanar distances and plane-to-plane angles through the Fourier transform of the images, the NC facets can be directly linked to the {221} and {310} planes. The (221) and (310) reflection rings are almost inseparable in the electron diffraction pattern, but the angles among them are critical in identifying the facets. From the images shown in Figure 9.11 (a, b), where the electron beam is parallel to [62 $\bar{1}$], the NCs are partially enclosed by the $\pm$(2 $\bar{1}$ 2), $\pm$(122), and ($\bar{1}$30) facets. When the electron beam is parallel to [001] [Figure 9.11(c)], the ($\bar{3}$ $\bar{1}$0) and ($\bar{1}$ 30) facets are imaged edge-on. Along the [120] direction [Figure 9.11(d)], the (2 $\overline{12}$) and (2 $\bar{1}$ 2) facets are imaged edge-on. These HRTEM images recorded from three different NC orientations are used to reconstruct the NC 3D shapes.

Figure 9.12(a) shows a polyhedral model of the nanocrystal enclosed by $\pm$(2 $\bar{1}$ 2), $\pm$(122), $\pm$($\bar{1}$ 30), $\pm$(310), $\pm$(2 $\overline{12}$), and $\pm$(1 2 $\bar{2}$) facets.[11] The structure can be simply constructed from a square based rod defined by

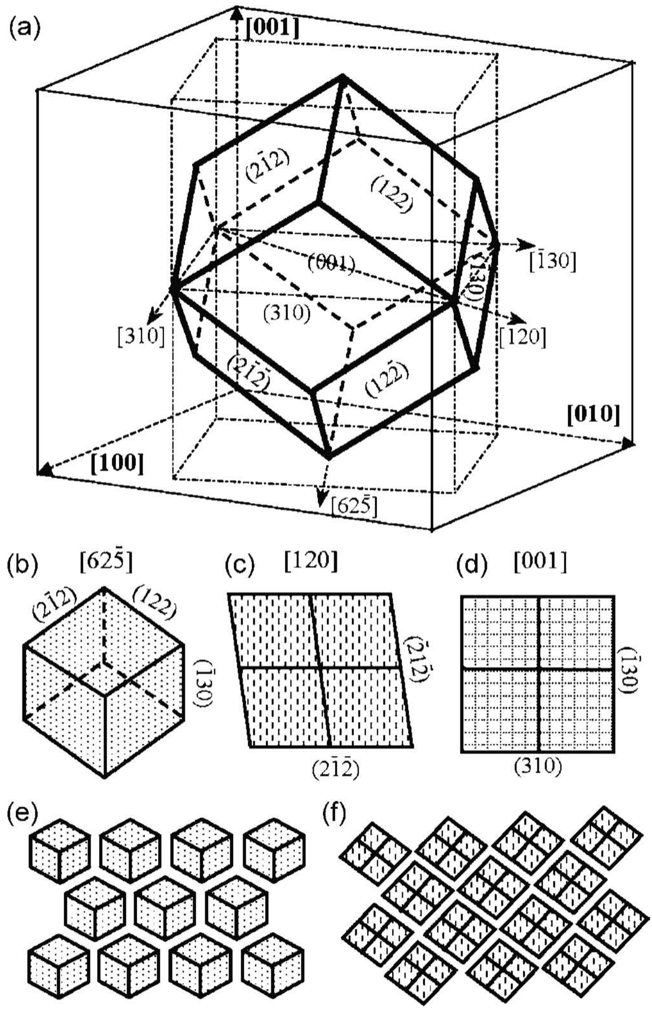

FIGURE 9.12 (a) Three-dimensional polyhedral shape of the Co nanocrystals constructed using the information provided by HRTEM images. (b–d) The projected shapes of the polyhedral model along [62 $\bar{5}$], [120], and [001], respectively. (e, f) Monolayer close-packing of the polyhedra oriented in [62 $\bar{5}$] and [120], respectively.

the ± ($\bar{1}$ 3 0) and ± (310) facets, which is cut by the (2 $\bar{1}$ 2), (122), ($\bar{2}$ 12), and ($\bar{1}$ $\bar{2}$ 2) facets at the top and by the ($\bar{2}$ 1 $\bar{2}$), ($\overline{122}$), (2 $\bar{1}$ $\bar{2}$), and (12 $\bar{2}$)facets at the bottom. The projected shapes of the polyhedron along [62 $\bar{5}$], [120], and [001] are given in Figure 9.12(b–d), respectively. These projections match well with the NC shapes observed in the images displayed in Figure 9.12. Using the model described here, the local orientation order in the self-assembly of Co NC has been successfully interpreted.

9.3.2 Twining Structure and Stacking Faults

Twining is one of the most common planar defects in NC, and it is frequently observed for fcc structured metallic NC possibly because it lowers the energy of the entire system. Twining is the result of two subgrains by sharing a common crystallographic plane, thus, the structure of one subgrain is the mirror reflection of the other by the twin plane. Figure 9.13(a) gives a HRTEM image of an FePt NC, showing the presence of a twin.

Stacking faults are typical planar defects. Stacking faults are produced by a distortion to the stacking sequence of atom planes. The (111) plane stacking sequence of a *fcc* structure follows A–B–C–A–B–C–A–B–C–. If the stacking sequence is changed to A–B–C–A–B–A–B–C–, a stacking fault is created.

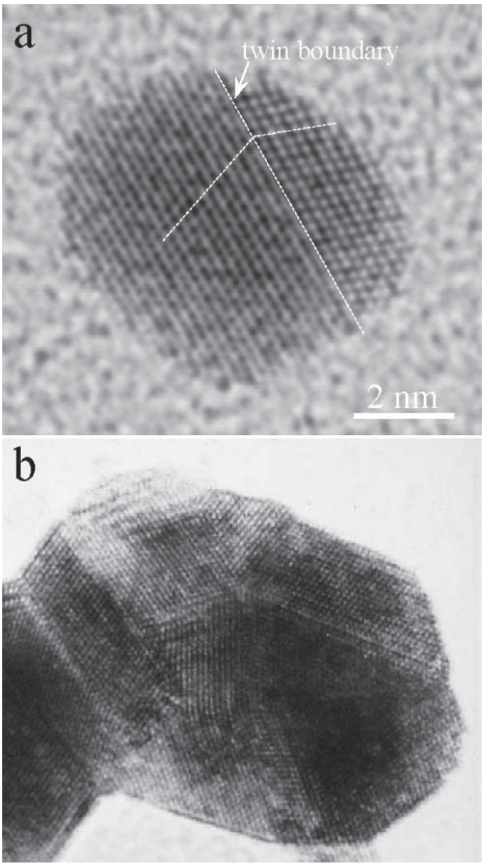

FIGURE 9.13 HRTEM images of (a) FePt and (b) Au nanocrystals having twin (T) and stacking fault (S) structures. The nanocrystals are oriented along [110] and the twin plane and stacking fault are parallel to the direction of the electron beam.

Figure 9.13(b) shows an Au particle that contains twins and stacking faults near the twins. It is known that NC usually contains strain but no dislocation. Twins and stacking faults are probably created due the high strain energy in the volume.

9.3.3 Multiply Twinned Particles—Decahedron and Icosahedron

The two most typical examples of multiple twinned particles (MTP) are decahedra and icosahedra.[12] Starting from an *fcc* structured tetrahedron, a decahedron is assembled from five tetrahedra sharing an edge [Figure 9.14(a)]. If the observation direction is along the five-fold axis and in an ideal situation, each tetrahedron shares an angle of 70.5°, meaning that five of them can only occupy a total of 352.6°, leaving a 7.4° gap. Therefore, strain must be introduced in the particle to fill the gap. An icosahedron is assembled using 20 tetrahedra via sharing an apex [Figure 9.14(b, c)]. The icosahedral and decahedral particles are the most extensively studied twinned NC.[13]

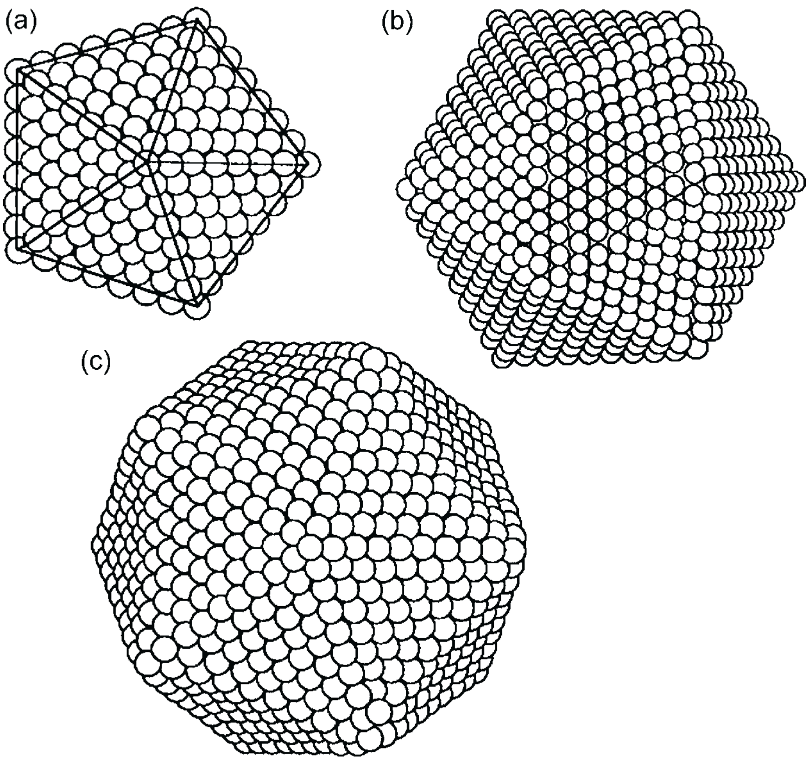

FIGURE 9.14 Atomic models of (a) decahedral, (b), and (c) icosahedral nanocrystals.

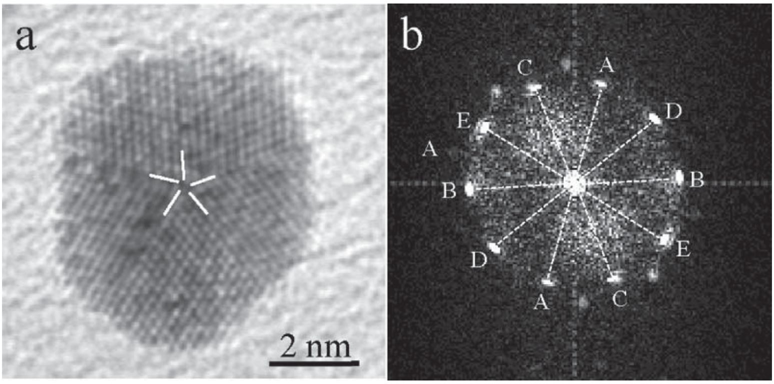

FIGURE 9.15 TEM images of decahedral FePt nanocrystals when the incident electron beam is parallel or nearly parallel to the fivefold symmetry axis. The strain field introduced by the five-fold twins is apparent in (a). (b) A Fourier transform of the image shown in (a) clearly reveals the fivefold symmetry.

The easiest orientation for identifying the MTPs is along the five-fold symmetry axis. Figure 9.15(a) shows a TEM image recorded from a FePt nanoparticle that is made of five twinned sub-particles. The Fourier transform of the image clearly indicates the five-fold pattern.

9.4 NANODIFFRACTION

Electron diffraction is a powerful tool for materials research ever since the invention of TEM. A popular experimental set up for electron diffraction is the so called selected area diffraction (SAD). Using parallel illumination, the area of interest is selected by placing an aperture in the image plane of the objective lens. Because of the physical size of the aperture, the lens aberration, and diffraction, the practical selected area by the aperture is limited to micrometer level. For conventional materials with a grain size of 10–100 μm, the diffraction pattern is a spot pattern. However, for nanostructured materials with a grain size of, say 20 nm, the diffraction pattern by the SAD method becomes a ring pattern (also called powder pattern) because so many grains are included in the aperture. Such a ring pattern contains rich information about the structure of the specimen and can be used to identify uniquely a known phase.[14] However, it is occasionally necessary to obtain spot pattern from a single grain of nanometer dimension for applications such as determination of crystal orientation, crystal structure of modulated phase, and epitaxy. This is the nanodiffraction technique, e.g. electron diffraction from a nanosized area of a specimen.

9.4.1 Optics for Nanodiffraction

Unlike SAD, nanodiffraction is realized by focusing the electron beam on the specimen. The experimental set up for nanodiffraction is schematically shown in Figure 9.16(b), in comparison with SAD in Figure 9.16(a). The area contributing to diffraction is controlled by the probe size. Since the probe size can be as small as 0.2 nm, nanodiffraction can therefore be readily achieved from an area smaller than the size of the unit cell. When the probe is focused, a minimum probe size can be obtained. The diffraction spots in the diffraction pattern are disks instead of sharp spots. The diameter of the disk is defined by the semi-convergence angle of the electron beam. This could cause overlapping of spots if the unit cell is large.

9.4.2 Experimental Procedure to Obtain a Nanodiffraction Pattern

The procedure for carrying out nanodiffraction is outlined below.[15]

1. Choose a proper probe size: a probe size smaller than half of the grain size under study is preferred. The smaller the probe size, the less chance there is to encounter a defect. However, if the probe is too small, the diffraction pattern would be too dark to be seen. Another point that needs to be mentioned is that the probe size must be a few times larger than the unit cell in order to preserve the symmetry of the crystal in the diffraction pattern.
2. Align the high voltage center and current center. Both are important for obtaining a good probe.
3. Correct astigmatism of the condenser lens in image mode by making the electron beam circular.
4. Correct astigmatism of the objective lens. This can be done using a larger spot size. This alignment ensures that a minimum spot size can be obtained.
5. Focus the image. This is done by setting the objective current at a desired value through adjusting the z control (the vertical translation of the specimen). Fine focus is done by adjusting the objective current.

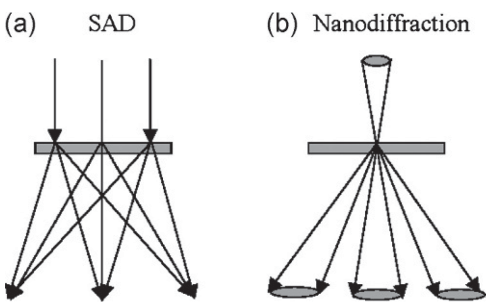

FIGURE 9.16 Schematic illustration of the experimental setup for (a) SAD and (b) nanodiffraction.

6. Find the zone axis. Going through the above procedure will generate a good probe. Now going to the diffraction mode will generate a diffraction pattern. Although tilting can still be used to find the proper zone axis, large angle tilting is limited by the size of the grain and can cause overlapping with adjacent grains. Therefore, finding the zone axis is done by alternately probing a grain while watching the diffraction pattern and making appropriate adjustments to specimen position and orientation until a grain with the desired zone axis is found.
7. Expose the film. A shorter exposure time of 1–2 seconds is preferred. This can be done by using a smaller camera length.

The trick to taking good quality nanodiffraction patterns is to reduce the beam illumination time on the crystal under examination. A nanoprobe placed on crystals can cause very fast contamination. The contaminants are light elements such as carbon. The clear diffraction pattern can disappear in a few seconds. The contamination can come from poor vacuum or from the specimen surface. Using a cold specimen stage is an effective way to reduce specimen contamination. The contaminant atoms can be "frozen" at the specimen surface. Heating the specimen to 273 K (100 °C) is also occasionally helpful to eliminate contaminants from the specimen surface.

9.5 SCANNING TRANSMISSION ELECTRON MICROSCOPY

A dedicated scanning transmission electron microscope (STEM) is related to the TEM via the principle of reciprocity in optics: given an imaging system consisting of a source, a series of lenses and apertures, and a detector, the same image can be obtained by interchanging the positions of the source and the detector (Figure 9.17). (The term "dedicated" is used to differentiate between an electron microscope designed primarily to be used as a STEM from a TEM with a STEM mode.) In a TEM, the electron source is typically regarded as a point source, which corresponds to a point detector or a "point image" in a STEM. Therefore, in order to create a 2D image in a STEM, it is necessary to scan the beam across the specimen. The advantage of using a dedicated STEM in nanoscale studies of materials lies in the very small probe that is formed by the gun lens—condenser lens—objective lens/aperture combination and the high brightness of the electron source.

9.5.1 Instrumentation

Because the STEM and TEM are related by the principle of reciprocity, many of the lenses and apertures used in TEM retain their positions relative to the specimen, although in the STEM they may serve a different function (see Figure 9.17). One key difference is the scan coils located prior to the beam's entry into the objective lens in STEM that is used for rastering the beam across

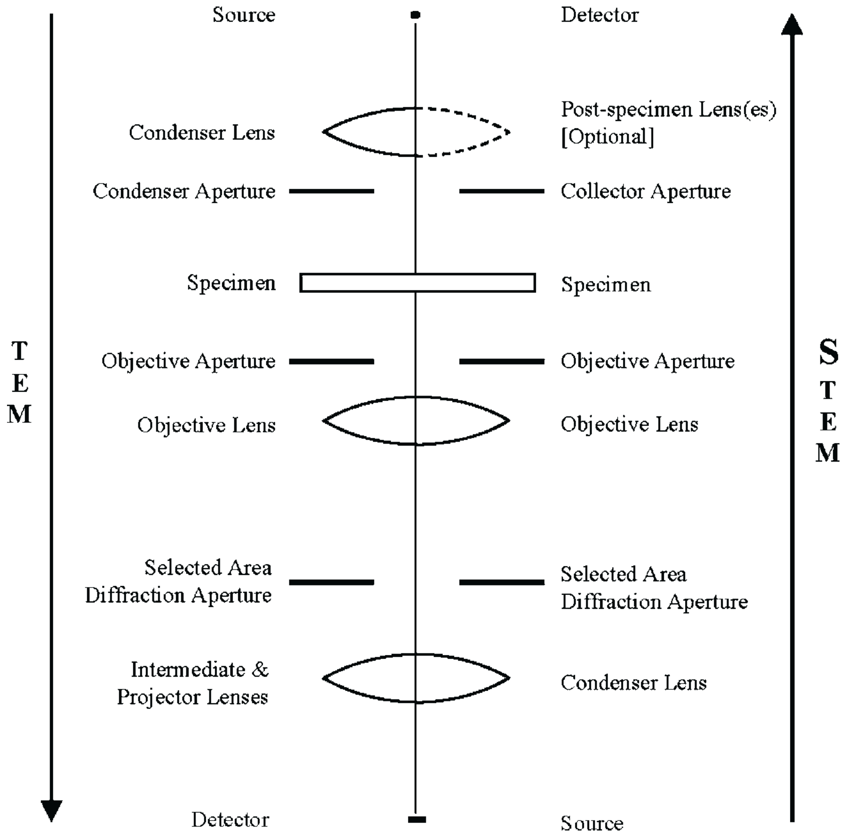

FIGURE 9.17 Schematic diagram showing the relationship between STEM and TEM. The sequence of lenses and apertures traversed by an electron in a TEM is shown on the left-hand side of the diagram, reading downward. An electron in a STEM sees the sequence of lenses and apertures indicated on the right-hand side, reading upward (note that some STEMs do not have post-specimen lenses). The optic axis is shown as a vertical line in the center of the diagram.

the specimen. The small beam diameter produced in STEM necessitates a high brightness electron source for sufficient signal. Field emission sources such as crystalline tungsten tips are frequently used.

Another major difference between a dedicated STEM and a TEM lies in the electron detectors (and therefore imaging modes) that are typically used. In a TEM, the objective aperture determines whether a BF or DF image is acquired and usually the same detector is used for acquiring both types of images. This implies that BF and DF images cannot be obtained simultaneously in TEM. In a typical dedicated STEM, however, there are separate BF and annular DF (ADF) detectors (Figure 9.18). This allows for the simultaneous acquisition of BF and ADF images from the specimen with each image containing different information about the specimen.

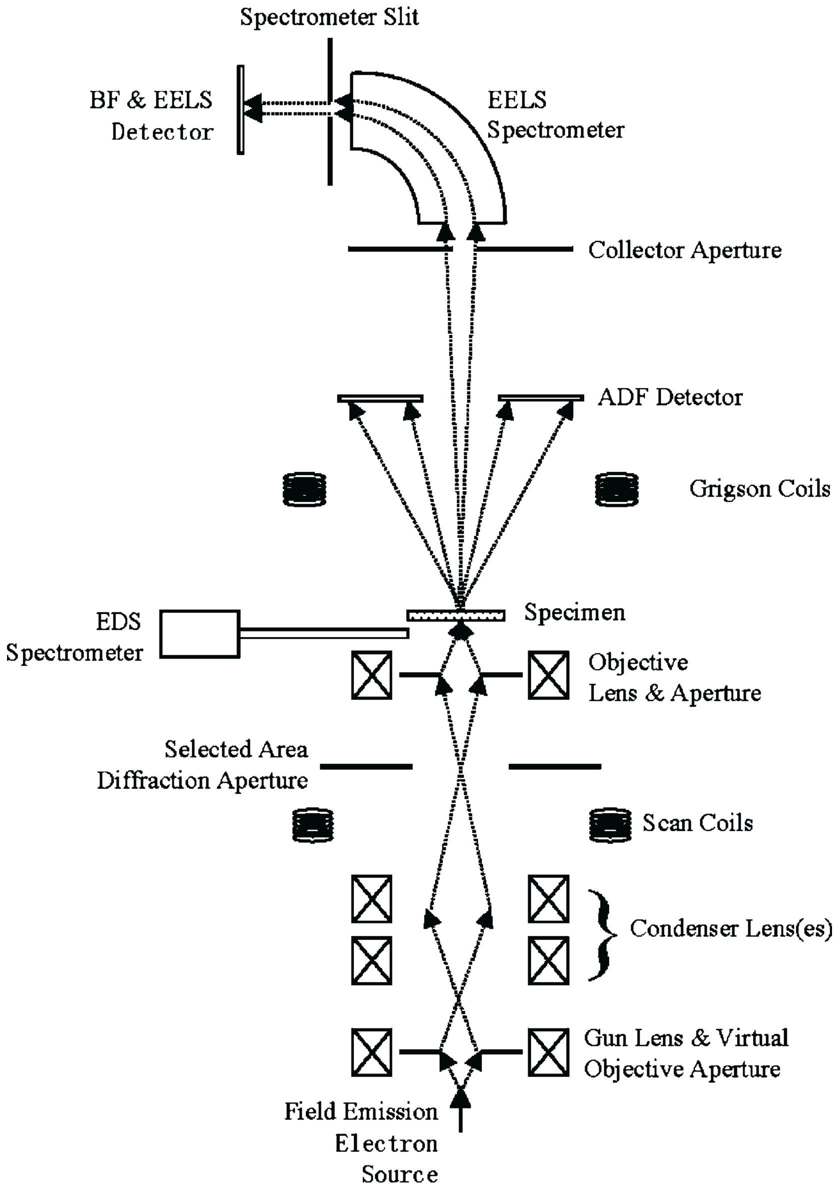

FIGURE 9.18 Schematic diagram showing a typical dedicated analytical STEM. Dotted arrows show examples of electron trajectories in the STEM.

Spectrometers and detectors for analytical work (e.g. EDS and EELS) can also be integrated into a dedicated STEM in a manner similar to that used in conventional TEM microanalysis.

9.5.2 Imaging Modes

Both BF and DF imaging modes similar to that in TEM are available in STEM. Electrons that are unscattered by the specimen or are scattered through a small angle (a few mrad) by the specimen can pass through the hole in the center of the ADF detector and proceed on to the BF detector. A collector aperture located ahead of the BF detector serves a purpose analogous to the objective aperture in TEM by limiting the angular spread of the electrons contributing to the BF image. The hole in the ADF detector itself can also be used as a collector aperture for BF imaging. On-axis "conventional" DF images can be obtained from the BF detector in STEM by using a post-specimen set of coils (Grigson coils) to deflect the 000 beam away from the optic axis and scan the desired scattered beams onto the BF detector.

The position of the objective aperture prior to the beam's entry into the specimen and the separation of the BF and ADF detectors in a STEM also gives rise to a different method of obtaining DF images than that used in TEM. In TEM, the specimen is located between the electron source and the objective aperture, and one of the primary functions of the objective aperture is to exclude the 000 beam while selecting the desired diffracted beam(s) to be used in creating a conventional DF image. However, in STEM, the objective aperture is located between the electron source and the specimen, and it plays a key role in determining the size of the probe incident upon the specimen. The ADF detector typically receives electrons that have undergone large angle scattering ($\sim$50–200 mrad) from the specimen; a plot of the probe position on the specimen vs. the intensity that is read from this detector is the ADF image.

Unlike TEM DF images that are usually formed using a small sector of reciprocal space, ADF images incorporate electrons scattered by the specimen into a cone covering the entire 360° polar angle range and a range of azimuthal angle that depends on the angular extent and location of the detector. A mask can be placed over part of the ADF detector to obtain the desired azimuthal angle range. Electrons scattered into the region intercepted by the ADF detector include atomic number-sensitive scattering from the specimen, specimen defects, strain, phonons, and intensity from Laue zones. Thus a complete interpretation of ADF images takes into account these effects.

The physical separation of the BF and ADF detectors leads to another advantageous feature of STEM: BF and ADF images can be obtained simultaneously from a sample. This feature is extremely useful in cases where information from BF and ADF imaging is required from exactly the same location on the specimen, which is often the case in nanoscale studies of materials. Moreover, complementary information can often be obtained from the simultaneous BF and

ADF images of a specimen. For example, ADF images tend to be more sensitive to differences in atomic number and less prone to thickness-dependent contrast reversal than BF images. Because diffraction effects are significant in ADF images, a BF/ADF image pair can reveal specimen features such as small amorphous regions. Therefore, BF/ADF image pairs allow for a wealth of information to be extracted from a sample.

9.5.3 Composition-Sensitive Imaging

Two of the most common analytical techniques available on STEMs and TEMs are EELS and EDS; both of these methods are able to produce composition maps of specimens using STEM. In a STEM, an EEL spectrometer is located on the optic axis between the ADF detector and the BF detector. Incident electrons scattered by electrons in the specimen can undergo a characteristic energy loss that is derived from the difference between the binding energies associated with different shells in a given element. The scattered electrons are discriminated according to their energy loss by a bending magnet in the EEL spectrometer that acts on the electrons in a manner analogous to the way a prism acts on light. Electrons having a desired value of energy loss can be directed into the EEL spectrometer through the application of an offset voltage or by changing the current in the spectrometer magnet. The energy resolution of the EELS spectrum is determined by the width of the spectrometer slit and/or by the properties of the detector (such as in the case where a CCD detector is used). By selecting the appropriate value of energy loss and energy resolution, electrons having an energy loss characteristic of particular element are received by the detector and an image can be formed by plotting the intensity of the EELS signal as a function of probe position on the sample. This type of image is called an energy filtered image, where areas of high intensity on the image correspond to relatively large concentrations of the particular element. Quantitative composition interpretation of an energy filtered image depends on knowledge of the scattering cross-section of the relevant EELS edge, the sample thickness, and the energy resolution.

STEMs equipped with EDS and the appropriate software are also capable of producing composition maps. Using a strategy similar to energy filtered imaging, software designed to produce EDS composition maps plot the intensity of a characteristic X-ray peak as a function of probe position to map out the concentration of a specific element in a sample. A phenomenon called coherent bremsstrahlung is a type of channeling that can occur when electrons in the incident beam enter a crystalline sample. Depending on the incident electron energy and the incident angle, electrons can interact with the periodic structure of the sample to produce spurious X-rays peaks in the EDS spectrum.[16]

Because the diameter of the STEM probe is on the order of interatomic spacing, the probe may channel between columns of atoms in a crystalline sample if the sample is oriented in an appropriate manner with respect to the electron beam. A method called ALCHEMI[17] takes advantage of the orientation dependence of an atom's ability to become ionized in a crystalline sample. Since characteristic X-rays are often emitted when an ionized atom seeks to lower its energy, the intensity of a given characteristic X-ray peak is expected to vary as a function of sample orientation. Therefore, ALCHEMI represents another way in which X-rays can be used to give localized composition information about a sample.

9.5.4 Nanoscale Microanalysis

As mentioned earlier, the orientation of crystalline samples in STEM can be an important factor in acquiring nanoscale analytical information using a technique such as ALCHEMI. Because STEM and TEM examine samples in projection, it is often also beneficial to use very thin samples for STEM or TEM to avoid any "overlap" effects that can occur when one nanoparticle is located directly above another nanoparticle along the beam direction. Moreover, thinner specimens reduce the likelihood of multiple scattering events and subsequently the beam broadening that can diminish analysis spatial resolution.

Given this preference for thin samples, EELS-based analytical methods tend to be favored over EDS-based analytical methods for studies of nanomaterials. Among the reasons for this are: (1) the generation of electrons with characteristic energy loss is more efficient than the generation of characteristic X-rays and (2) the interpretation of EELS data is considerably simplified when the sample thickness is less than the mean free path of the incident electron in the sample.[18] Nevertheless, the availability of more than one analytical method on an electron microscope can be very useful for discerning composition information from samples that contain elements whose characteristic edges overlap in a given analytical method or where detection of certain elements is difficult or impossible. An example of this is the inability of EDS to detect elements having atomic number $Z < 5$; EELS is able to detect these elements as long as they are present in the specimen in sufficient quantities.

9.6 IN-SITU TEM AND NANOMEASUREMENTS

The presence of a large percentage of surface atoms in nanomaterials results in surface dominated phenomena. The large mobility of the surface atoms and the faceted shapes of the NC that are not formed under thermodynamic equilibrium conditions could have very interesting physical, chemical and electronic structures of fundamental interests and technological importance. TEM is an ideal tool for tracing the in-situ behavior of NC induced by temperature and/or externally applied fields. The in-situ experiments can be carried out on the same nanocrystal over a large range of temperature, giving the possibility of examining the temperature induced shape transformation, structural evolution, melting and surface phenomena. This section focuses on the several types of in-situ TEM techniques for probing the unique properties of NC.

9.6.1 Thermodynamic Properties of Nanocrystals

The thermodynamic properties of NC depend strongly on not only the crystal size but also their shapes.[19] The thermodynamic properties of Au nanorods are an example. A series of in-situ TEM images recorded for the same specimen region as the temperature was increased continuously from room temperature to 923 K show the formation process of Au nanoparticles on the carbon support in Figure 9.19(a–h). The nucleation started at 473 K as shown in Figure 9.19(b) and it was only observed by TEM at 488 K shown in Figure 9.19(c). It is clear that as the temperature reached 433 K the structure of the micelles around the rods was found to collapse in solution and no visible change was observed in the morphology of the nanorods. At $T=488$ K [Figure 9.19(c)], small nuclei of gold nanoparticles began to form on the surface of the carbon substrate. A small increase in temperature to 493 K [Figure 9.19(d)] resulted in a rapid growth of these small particles on the carbon substrate. Further increase in temperature to 513 K led to the growth of the particles, and the stable particle size was reached at $T=600$ K. At $T>513$ K, the widths of the nanorods increased significantly, while their lengths remained the same [compare the rod marked by

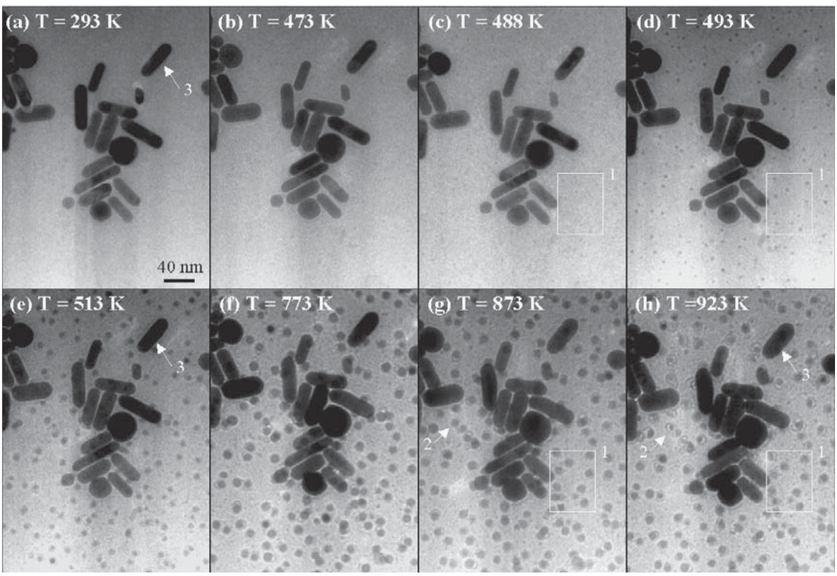

FIGURE 9.19 (a–f) show the result of the sublimation, condensation, and diffusion of gold atoms from gold nanorods. A series of in-situ TEM images recorded for the same specimen region as the temperature was increased continuously from room temperature to 923 K, showing the formation process of Au small nanoparticles on the carbon support. The nucleation started at ~473 K and it was only clearly observed by the TEM at 488 K shown in (b). At $T>513$ K, the widths of the nanorods increased significantly, while their lengths remained the same (compare the rod marked by an arrow in (a), (e), and (f)). This is due to the surface melting which induces shape transformation, so that the thickness of the nanorods decreased when placed onto the substrate as a "liquid-like" material (note an increase in the width does not necessarily mean an increase in the volume).

an arrow in Figure 9.19(a, e, h)]. This is due to surface melting that induces shape transformation, so that the thickness of the nanorods decreased when placed onto the substrate as a "liquid-like" material. It is noticed that the density of the particles in the regions adjacent to the Au rods is approximately the same as the particles in the regions without rods.

We propose that the formation of small Au nanoparticles in Figure 9.19 (d–h) is the result of surface sublimation from gold nanorods as well as the gold atoms contained in the solution deposited onto the carbon substrate. The sublimation of gold atoms from the surface is supported by following observation. Shown in Figure 9.20 is a series of TEM images recorded from the same area in which the shrinkage of a particle indicated by an arrowhead is visible. Such a drastic reduction in size could be the result of surface sublimation, and more interestingly, this phenomenon occurs at a relatively low temperature.

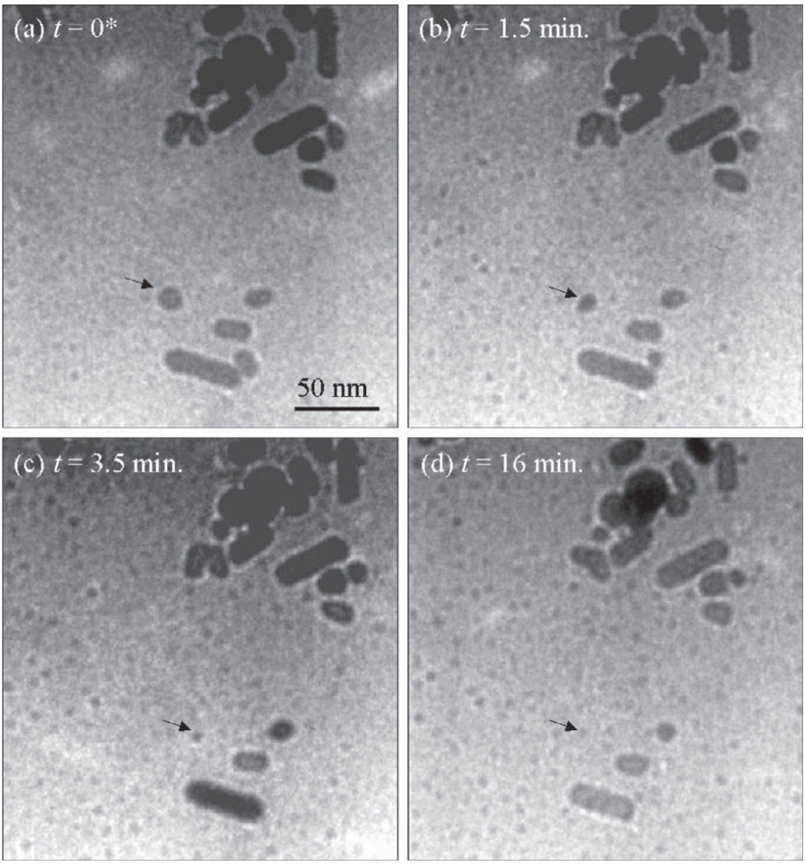

FIGURE 9.20 In-situ TEM images recorded as a function of time at 498 K (225 °C), showing the shrinking of an Au particle indicated by an arrowhead at such a low temperature, indicating surface sublimation and diffusion. (a) $t=0$; (b) $t=1.5$ min; (c) $t=3.5$ min; (d) $t=16$ min.

A model can be proposed to understand the observed phenomena. The initial rapid growth of the Au particles is due to the contribution of the diffusion of atoms on the substrate surface. This process is assumed to follow the hit- and -stick model and is directly related to the intersection length of the particle with the substrate surface. On the other hand, the atoms on the surface of the Au particles can sublime, and their sublimation rate is directly proportional to the exposed surface area of the particle. Accordingly, for a spherical particle of radius r, partially embedded in the substrate as shown in Figure 9.21(c), the net growth of the number of atoms (n_r) in the particle is given by:

$$\frac{dn_r}{dt} = (2\pi r \sin\theta) N_0 k_d - 4\pi r^2 \alpha \rho_S k_S \tag{9.17}$$

where θ is the angle that specifies the degree of contact of the particle with the substrate; k_d is the diffusion rate constant of the atoms adsorbed on the substrate; N_0 is the steady state concentration of the Au atoms on the surface; α is a factor

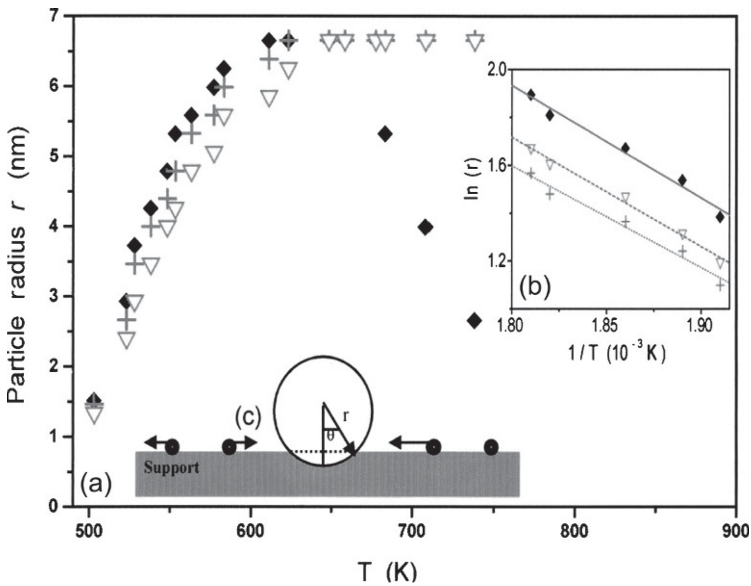

FIGURE 9.21 (a) The dependence of the steady state particle radii (r_{max}) formed at different specimen temperatures for particles located at different distances from the Au rods (diamond shapes are for the closest to the rod and triangles are for the furthest away). The size of the particles located closest to the rod (~50 nm) drops sharply at 650 K. Considering the specimen drift at higher temperature and the effect of carbon substrate on the image contrast, the newly formed particles were seen in TEM only when their sizes were larger than ~2 nm. ◆, +, and ∇ represent the particles at distances of 50, 100, and 200 nm, respectively, from the gold nanorods. (b) The ln(r_{max}) versus $1/T$ plot in the low-temperature range, the slopes of which gives the difference between the activation energy of diffusion and the sublimation energy Q of the gold atoms. (c) A sketch showing the growth process of the small gold cluster on the TEM substrate. Growth kinetics of the small particles at a constant temperature of $T = 498$ K.

representing the fraction of the exposed surface area; ρ_s is the surface density of the particle; and k_s is the sublimation rate constant. The direct condensation of Au atoms onto the particle surface can be accounted for by a proper choice of the α factor. Initially, the diffusion term is dominant, resulting in a rapid growth of the particle. As the particle grows in size, the sublimation term increases because of the r^2 term. When the two terms are equal, a steady state is reached ($dn_r/dt=0$), achieving a maximum radius ($r_{\max}$) at this temperature, which is given by

$$r_{\max} = \frac{N_0 k_d \sin\theta}{2\alpha \rho_S k_S} \tag{9.18}$$

Using the relation $nr = 4/3\ \alpha r^3 \rho_0$, where ρ_0 is the volume density of Au atoms, and the boundary condition $r=0$ at $t=0$, the solution of Eq. (9.17) relating r to t is given by:

$$r_{\max} \ln\left(\frac{r_{\max}}{r_{\max} - r}\right) - r = [\alpha \rho_S k_S / \rho_0] t \tag{9.19}$$

The kinetics of Au particle growth was determined using in-situ TEM at 498 K, as soon as the newly formed particles were observed in TEM. From the kinetic curve given in Figure 9.22(a), the particles seem to experience a rapid growth in the first 500 s, and then the size reached a constant value.

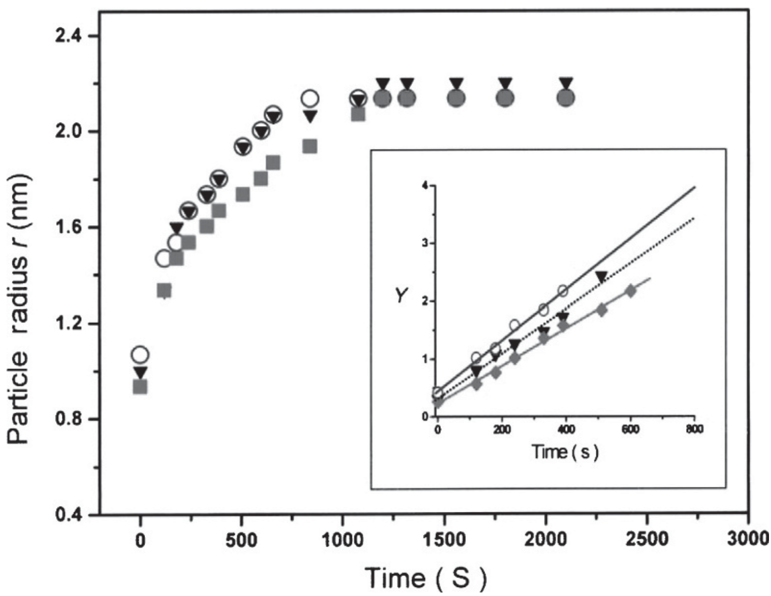

FIGURE 9.22 (a) Shows the particle radii as a function of time. (b) Shows a comparison of the theoretically calculated Y versus time curves according to Eq. (9.19) with the experimentally observed data points. From the slopes of the lines, the sublimation energy Q is calculated.

By setting the left-hand side of Eq. (9.19) equal to Y and using the experimental data in Figure 9.22(a), the plot of Y versus time can be used to calculate the activation energy Q for sublimation of gold atoms, which is $Q = 1.39 \pm 0.01$ eV.

The crystallographic shape of the Au nanorods has been investigated using HRTEM to understand why the gold atoms can sublimate at lower temperature. The nanorods have been found to exhibit the unstable {110} facets (Figure 9.23), in addition to the {100} and {111} facets present in the spheres.[20] For the face-centered Au NC, the surface energy follows a relationship of $\gamma\{110\} > \gamma\{100\} > \gamma\{111\}$. The presence of a significant percentage of gold atoms on the {110} surface suggests that the nanorod is relatively thermodynamically unstable. The {111} and {100} faceted Au spheres have lower surface energy. Thus, their atoms may not sublime at relatively low temperatures. Therefore, the sublimation of Au atoms from the nanorods at lower temperatures is due to the shape of the nanorods, which has unstable {110} surfaces.

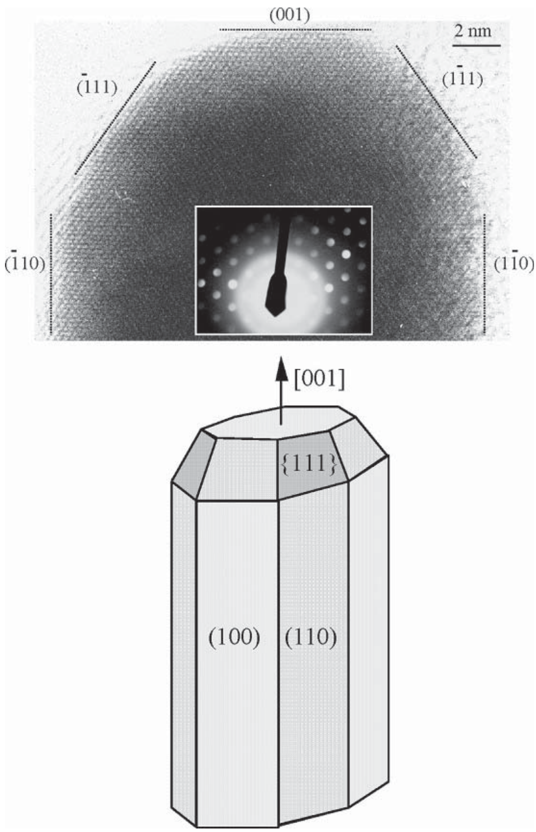

FIGURE 9.23 HRTEM image showing the {110} and {100} facets of the Au nanorods. The electron beam is parallel to [110].

Study of shape-controlled NC has a great interest from a catalysis viewpoint. Platinum nanoparticles with a high percentage of cubic-, tetrahedral-, and octahedral-like shapes, respectively, have been synthesized by changing the ratio of the concentration of polymer capping material (polyacrylate) to that of Pt^{2+} being reduced by H_2 from K_2PtCl_4 at room temperature.[21] The polymer acts not only as the passivative protection layer of the NC, but it is also the key factor in controlling their shapes.[22,23] The growth mechanism of the shape controlled Pt NC is attributed to a kinetic process in which most of the nuclei are tetrahedral, while a shape transformation from tetrahedral to cubic occurs as the particles grow.[24] The in-situ behavior of the Pt NC has also been studied by TEM.[25] The surface capping polymer was removed by annealing the specimen to a temperature of 453–523 K (180–250 °C), while the particle shape showed no change up to ~623 K (350 °C). In a temperature range of 623–723 K (350–450 °C), a small truncation occurred in the particle shape but no major shape transformation. The particle shape experienced a dramatic transformation into a spherical-like shape when the temperature was higher than ~773 K (500 °C); the macroscopic surface melting occurred at ~873 K (600 °C), much lower than the melting point of bulk Pt (2046 K or 1773 °C).

9.7 ELECTRON ENERGY-LOSS SPECTROSCOPY OF NANOPARTICLES

Chemical microanalysis is an important field in TEM. Under the impact of an incident electron beam, the electrons bounded to the atoms in the specimen may be excited either to a free electron state or to an unoccupied energy level with a higher energy. The quantum transitions associated with these excitations will emit photons (or X-rays) and electrons such as SEs, Auger electrons, and ionized electrons. These inelastic scattering signals are the fingerprints of the elements that can provide quantitative chemical and electronic structural information.

9.7.1 Quantitative Nanoanalysis

Energy dispersive X-ray spectroscopy[1] and EELS[18] in TEM have been demonstrated as powerful techniques for performing microanalysis and for studying the electronic structure of materials. Atomic inner shell excitations are often seen in EELS spectra due to a process in which an atom-bound electron is excited from an inner shell state to a valence state accompanied by incident electron energy loss and momentum transfer. This is a localized inelastic scattering process, which occurs only when the incident electrons are propagating in the crystal. Figure 9.24 shows an EELS spectrum acquired from a superconducting compound $La_{0.5}Sr_{0.5}CoO_3$, where the ionization edges of O K, Co–L, and La–M are present in the displayed energy-loss range. Since the inner-shell energy levels are the unique features of the element, the intensities of the

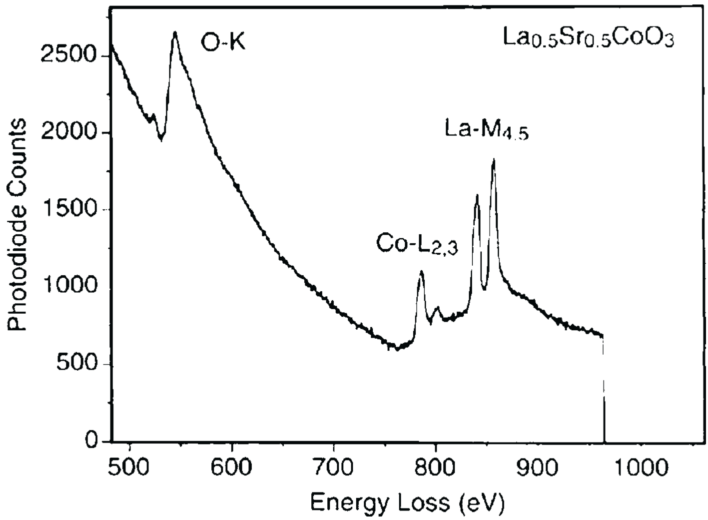

FIGURE 9.24 An EELS spectrum acquired from $La_{0.5}Sr_{0.5}CoO_3$ showing the application of EELS for quantitative chemical analysis.

ionization edges can be used effectively to analyze the chemistry of the specimen.

After subtracting the background, integration is performed to the ionization edge for an energy window of width Δ that takes into account the threshold energy. Thus, the intensity of oscillation at the near edge region is flattened, and the integrated intensity is dominated by the properties of single atoms. This type of information is most useful in material analysis and the integrated intensity is given by

$$I_A(\Delta) \approx I_0(\Delta)\sigma_A(\Delta,\beta)n_A d \tag{9.20}$$

where $I_0(\Delta)$ is the integrated intensity of the low-loss region including the zero-loss peak for an energy window Δ; $\sigma_A(\Delta,\beta)$ is the energy and angular integrated ionization cross-section. In imaging mode, β is mainly determined by the size of the objective aperture or the upper cut-off angle depending on which is smaller. In diffraction mode, the β angle is determined not only by the size of the EELS entrance aperture and the camera length but also by the beam convergence. In general, the width of the energy window is required to be more than 50 eV to ensure the validity of Eq. (9.20), and $\Delta = 100$ eV is an optimum choice. If the ionization edges of two elements are observed in the same spectrum, the chemical composition of the specimen is

$$\frac{n_A}{n_B} = \frac{I_A(\Delta)\sigma_B(\Delta,\beta)}{I_B(\Delta)\sigma_A(\Delta,\beta)} \tag{9.21}$$

This is the most powerful application of EELS because the spatial resolution is almost entirely determined by the size of the electron probe. The key quantity in this analysis is the ionization cross-section. For elements with atomic numbers smaller than 14, the *K* edge ionization cross-section can be calculated using the SIGMAK program,[18] in which the atomic wave function is approximated by a single-electron hydrogen-like model. The ionization cross-section for elements with $13 < Z < 28$ can be calculated using the SIGMAL program.[19] For a general case, the ionization cross-section may need to be measured experimentally using a standard specimen with known chemical composition.

9.7.2 Near Edge Fine Structure and Bonding in Transition Metal Oxides

The energy-loss near edge structure (ELNES) is sensitive to the crystal structure. This is a unique characteristic of EELS and in some cases it can serve as a "fingerprint" to identify a compound. In EELS, the L ionization edges of transition-metal and rare-earth elements usually display sharp peaks at the near edge region (Figure 9.25), which are known as *white lines*. For transition metals with unoccupied 3d states, the transition of an electron from 2p state to 3d levels leads to the formation of white lines. The L_3 and L_2 lines are the transitions from $2p^{3/2}$ to $3d^{3/2}3d^{5/2}$ and from $2p^{1/2}$ to $3d^{3/2}$, respectively, and their intensities are related to the unoccupied states in the 3d bands. Numerous EELS experiments have shown that a change in valence state of cations introduces a dramatic change in the ratio of the white lines, leading to the possibility of identifying the occupation number of the 3d orbital using EELS.

EELS analysis of the valence state is carried out in reference to the spectra acquired from standard specimens with known cation valence states. Since the

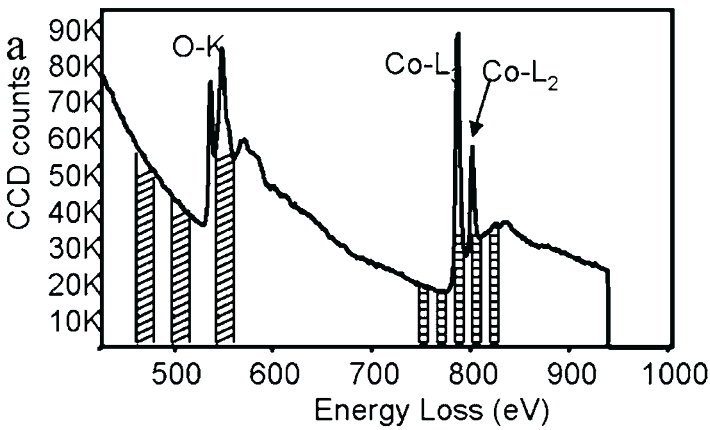

FIGURE 9.25 EELS spectrum acquired from cobalt oxide showing the Co-L_3 and Co-L_3 white lines. The five windows pasted in the Co-L edge are to be used for extracting the image formed by the ratio of white lines.

intensity ratio of L_3/L_2 is sensitive to the valence state of the corresponding element, if a series of EELS spectra are acquired from several standard specimens with known valence states, an empirical plot of these data serves as the reference for determining the valence state of the element present in a new compound.[26–29] The L_3/L_2 ratios for a few standard Co compounds are plotted in Figure 9.26(a). EELS spectra of Co-$L_{2,3}$ ionization edges were acquired from $CoSi_2$ (with Co^{4+}), Co_3O_4 (with $Co^{2.67+}$), $CoCO_3$ (with Co^{2+}) and $CoSO_4$

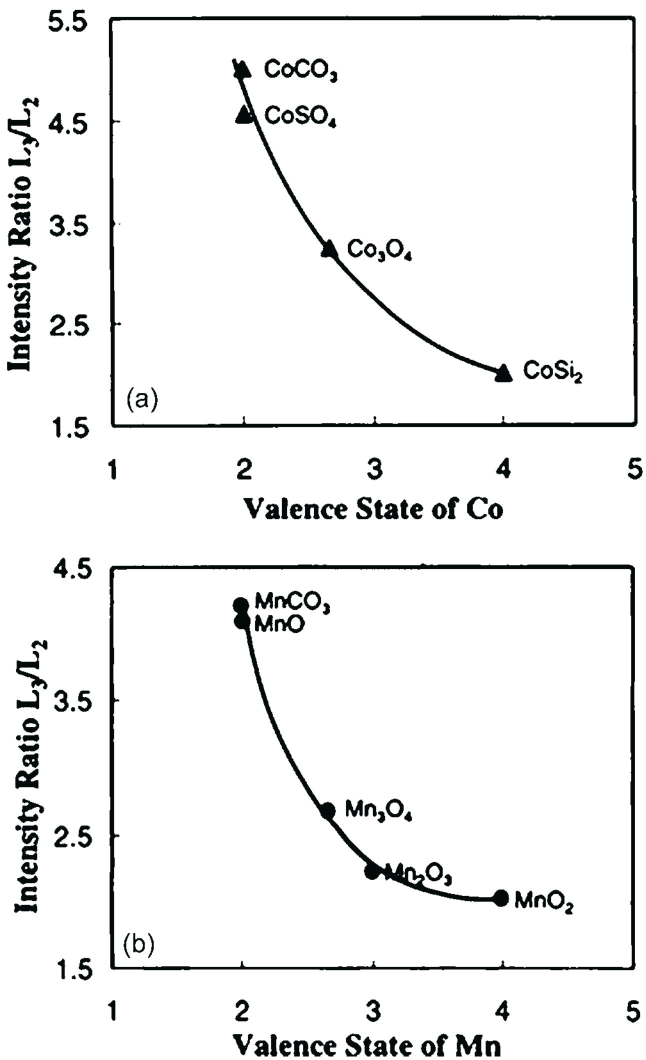

FIGURE 9.26 Plots of the intensity ratios of L_3/L_2 calculated from the spectra acquired from (a) Co compounds and (b) Mn compounds as a function of the cation valence. A nominal fit of the experimental data is shown by a solid curve.

(with Co^{2+}). This is the basis of our experimental approach for measuring the valence states of Co or Mn in a new material.

Determining the crystal structure of nanoparticles is a challenge particularly when the particles are smaller than 5 nm. The intensity maxima observed in the X-ray or electron diffraction patterns of such small particles are broadened due to the crystal shape factor, and greatly reduce the accuracy of structure refinement. The quality of the high-resolution TEM images of the particles is degraded because of the strong effect from the substrate. This difficulty arose in our recent study of CoO NC whose shape is dominated by tetrahedra of sizes smaller than 5 nm.[30] Electron diffraction indicates the crystal has an fcc-type cubic structure. The synthesized NC are likely to be CoO. EELS is used to measure the valence state of Co. Figure 9.27 shows a comparison of the spectra acquired from Co_3O_4 and CoO standard specimens and the synthesized NC. The relative intensity of the Co-L_2 to Co-L_3 for the NC is almost identical to that for CoO standard, while the Co-L_2 line of Co_3O_4 is significantly higher, indicating that the Co valence in the NC is 2+, confirming the CoO structure of the NC.

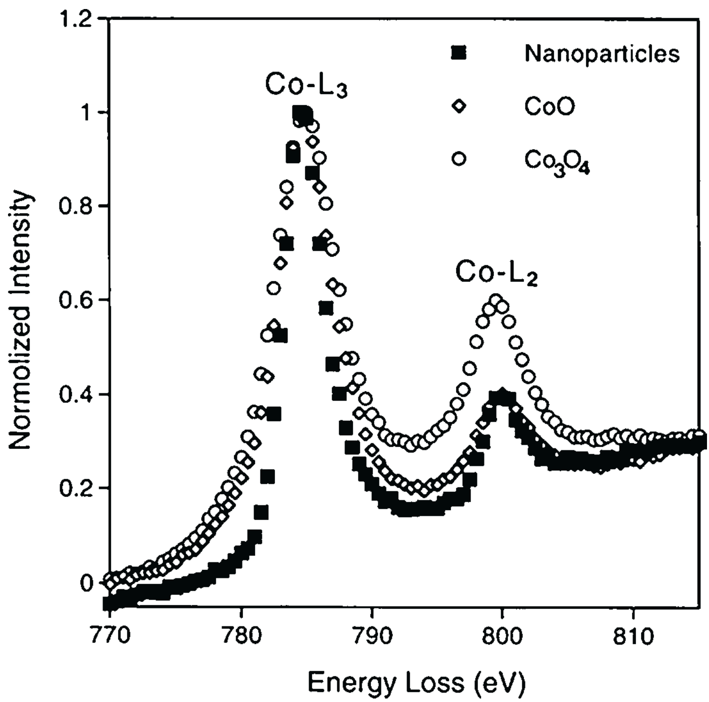

FIGURE 9.27 Comparison of EELS spectra of Co-L_2 and Co-L_3 ionization edges acquired from Co_3O_4 and CoO standard specimens and the synthesized nanocrystals, proving that the valence state of Co is 2+ in the nanocrystals. The full width at half maximum of the white lines for the Co_3O_4 and CoO standards is wider than that for the nanocrystals due to size effect.

LIVE GRAPH
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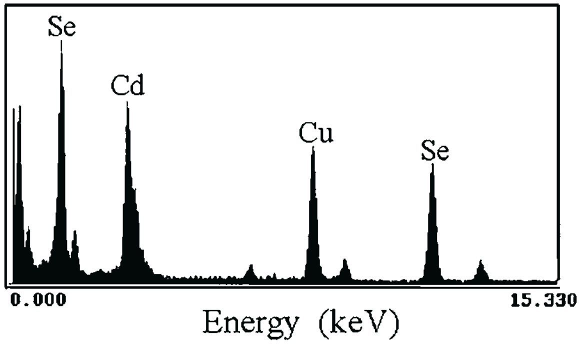

FIGURE 9.28 EDS spectrum acquired from a specimen of CdSe nanosaws. The copper signal came from the TEM grid.

9.8 ENERGY DISPERSIVE X-RAY MICROANALYSIS (EDS)

X-rays emitted from atoms represent the characteristics of the elements and their intensity distribution represents the thickness-projected atom densities in the specimen. This is the basis for EDS which has played an important role in microanalysis, particularly for heavier elements. EDS is a key tool for identifying the chemical composition of a specimen. A modern TEM is capable of producing a fine electron probe of smaller than 2 nm, allowing direct identification of the local composition of an individual nanocrystal. Shown in Figure 9.28 is an EDS spectrum acquired from a sample of CdSe nanosaws,[31] from which the chemical composition of the nanosaw structures is determined.

9.9 SUMMARY

There are several important developing directions in TEM-related techniques, including but not limited to (1) moving toward 0.1 nm image resolution using image processing or C_s corrector, (2) electron holography, (3) energy-filtered electron imaging and diffraction, (4) quantitative electron diffraction and imaging, (5) high-energy-resolution spectroscopy, (6) angstrom probe microscopy and spectroscopy, (7) environmental and in-situ microscopy, (8) in-situ nanoscale property measurements, and (9) analysis of real devices.

Transmission electron microscopy and associated techniques are powerful tools for characterization of nanophase materials. This chapter mainly introduced high-resolution imaging in TEM and some newly developed techniques, such as electron energy filtering. High-spatial resolution analysis is vitally important for solving many of the practical problems of nanomaterials. Spectroscopy analysis of solid state effects and valence states mapping are new directions of quantitative microscopy.

In-situ TEM is anticipated to be important for characterizing and measuring the properties of individual nanoparticles, from which the structure–property relationship can be clearly registered to a specific nanoparticle/structure. An emphasis has been placed on new developments in TEM for in-situ nanomeasurements of the mechanical and electrical properties of individual nanostructures, aiming to correlate the measured properties with the nanostructure. This is a new direction in TEM and it is important for characterizing nanophase materials. Therefore, TEM is truly a versatile tool not only for crystallographic and chemical structure analysis, but also a powerful approach for nanomeasurements.

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